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CFD-based model of adsorption columns: Validation

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ABSTRACT

The computational research on adsorption is of significant interest for process development and optimization. Despite the existence of unidimensional models that reproduce simple vertical columns, a further level of detail is required to understand the intricacy of transport phenomena within industrial-grade columns across time and space. With that in mind, this research work focuses on developing and validating a higher-dimensional model to reproduce the flow, heat, and mass transfer phenomena inside adsorption columns using Computational Fluid Dynamics (CFD). The mathematics of adsorption are coded into the commercial CFD solver, ANSYS Fluent v19.2, using User-Defined Functions (UDF). The model is implemented using a mathematical adaptation of Fluent's standard transport equations, so that the material and thermal effects of adsorption can be accounted for. The resulting model is validated against published experimental results for the single-component and competitive adsorption of N_2 and CO_2 on zeolite 13X on a small-scale pilot unit. Good agreement between our modelling results and experimental data is demonstrated. Afterwards, a modified column geometry with an inlet jet is used to explore the impact of gas injection on the unsteady spatial distribution of matter and energy. Radial gradients and 2D contour plots are illustrated in all cases to showcase the enhanced insight that a higher-dimensional model provides. Finally, this model can be used to predict comprehensive characteristics of adsorption columns of any design and size, taking into account different designs of inflow gas distributors in industrial scale columns. It includes a new sort of adsorption devices using monolith instead of packed beds. All model UDFs are open-source and can be found in supplementary materials.

1. Introduction

Gas adsorption plays a fundamental role within the realm of separation processes in engineering. It finds application in various fields, including but not limited to the concentration of oxygen from air (Jee et al., 2005a,b), hydrogen purification from steam methane reforming (Sircar and Golden, 2000; Majlan et al., 2009; Delgado et al., 2014), helium purification from natural gas (Jahromi et al., 2018a,b), and the removal of volatile organic compounds (Gales et al., 2003; Krishnamurthy et al., 2018; Roegiers and Denys, 2019). In addition, the prominent role of anthropological CO_2 emissions in accelerating climate change (Stern, 2008) has prompted several adsorption studies on carbon capture and sequestration (Zhang and Webley, 2008; Zhang et al., 2008; Grande et al., 2009; Haghpanah et al., 2013; Lian et al., 2019; Sandu et al., 2021).

This has facilitated developments in adsorbent candidate materials, pellet structure design, experimental investigations into adsorption cycle optimization, and –notably– advances in numerical methods and computational modelling. The reader is referred to the review articles by Shafeeyan et al. (2014), Ben-Mansour et al. (2016), and Akinola et al. (2022) that focus on prior mathematical modelling studies related to CO_2 adsorption from various mixtures as air, coke oven gas, cracked gas, flue gas, etc. Ben-Mansour et al. reflect on a plethora of CO_2 adsorption modelling studies. Of the 26 studies reviewed, all were 1D transient models, assumed ideal gas behaviour, and considered the Linear Driving Force (LDF) model for adsorption kinetics.

Due to the inherently transient nature of an adsorption process, it is crucial to understand the process dynamics within the packed column. One must consider coupled transport phenomena (material, momentum, and energy balances), thermodynamics, equilibrium isotherms, adsorption kinetics, and adsorbent properties to appropriately construct a mathematical model. Framing shell balances gives rise to a system of coupled partial differential equations (PDEs) distributed over time and space, along with auxiliary equations that describe adsorption equilibrium and many other process variables. Such systems often do not possess analytical solutions and, thus, need to be solved numerically. The

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Nomenclature

b	adsorption equilibrium constant for site 1
c	fluid phase molar concentration
c_p	specific heat capacity
d	adsorption equilibrium constant for site 2
D_m	molecular diffusivity
D_{eff}	effective diffusivity
D_L	dispersion coefficient
d_p	diameter of particle
E	energy
g	gravitational acceleration
H	enthalpy
K	permeability coefficient
k	LDF mass transfer coefficient
$k-\omega$	k-omega turbulence model
L	length of the domain
M	molar mass
RLX	relaxation factor
P	fluid phase pressure
Q_{iso}	isosteric adsorption heat
q	solid phase loading
$q*$	equilibrium solid phase loading
R	universal gas constant
R_{in}	column inner radius
R_{out}	column outer radius
R_W	column wall radius, equivalent to R_{in}
r	r-coordinate
r_p	radius of particle
T	fluid and solid phase temperature
t	physical time
V	volume
$\vec{v}$	velocity vector
Y	species mass fraction
y	species molar fraction
z	z-coordinate

Greek letters

ρ	density
ε	porosity or void fraction
κ_{eff}	effective thermal conductivity
Σ_v	diffusion volume
τ	tortuosity
μ	viscosity

Subscripts

a	adsorbed phase
i	species i
g	gas mixture, fluid phase
p	adsorbent particle

prominent choice for modelling transport phenomena has been 1D models. Equations are often coded and solved on programming platforms such as MATLAB, Fortran, PDECOL, DGEAR, ODEPACK, etc., where the Finite Volume Method (FVM) has been widely employed (Webley and He, 2000; Todd et al., 2001; Cruz et al., 2003, 2005; Javeed et al., 2011; Medi and Amanullah, 2011). In particular, it has been shown that 1D models can efficiently predict the behaviour of temperature and composition along the axial direction of an adsorption column for simple plug flow geometry, e.g., see the work (Haghpanah et al., 2013). However, it should be noted that 1D models are unable to predict the impact of inflow distributors on the heat and mass transfer inside large-scale industrial adsorption columns.

Computational Fluid Dynamics allows for a more detailed resolution of each transport equation down to a differential or 'cell' scale, in time and space. This is of particular importance when dealing with industrial-scaled columns, where the spatial effects of adsorption can become amplified by the increased size and complexity of the design. Here, a higher-dimensional or multidimensional adsorption model is required for a more accurate insight into phenomena such as entrance effects (due to an actual flow distributor), flow asymmetries (due to internals), dead zones, heat transfer gradients to the surroundings (including process modifications such as exterior column cooling), and even the effect of gravity within horizontal columns. Continuous improvements to achieve a better design, increased throughput, scalability, and techno-economic feasibility can thus benefit significantly from multidimensional modelling of adsorption processes.

Although commercial CFD packages include a module for simulating packed beds as porous media, they lack the required models or modules to adequately simulate adsorption within said packed beds, including equilibrium isotherms, adsorption kinetics, and transport terms to account for the effects of adsorption. This opens a research window: developing and implementing custom adsorption models based around CFD. Some attempts have been made, which are described next along with their achievements. The observed shortcomings and our attempt to address them are described afterwards.

Augier et al. carried out CFD simulations to explore hydrodynamic effects in both laboratory and industrial adsorbent columns (Augier et al., 2008). Industrial-scale simulations shed light on axial dispersion and flow perturbation inside the columns caused by trays, distributors, pipes and beams. They demonstrated that industrial-scale column hydrodynamics significantly impacted the adsorption yield, emphasizing the relevance of CFD-based research for the optimal design of internals in large-scale adsorption systems and fluid flow scenarios with high Reynolds numbers.

Verbruggen et al. applied a CFD-based approach to determine the adsorption parameters of acetaldehyde, a volatile organic compound, to abate indoor air pollution (Verbruggen et al., 2016). The Langmuir adsorption isotherm was enhanced by incorporating a chemical reaction mechanism, according to the Langmuir-Hinshelwood-Hougen-Watson (LHHW) framework. Transient simulations were performed on the COMSOL Multiphysics v.4.4 commercial software package. The CFD results and validation provided accurate acetaldehyde adsorption parameters for further photocatalytic kinetic analysis. This can be applied more widely to reactive adsorption systems.

Esposito et al. modelled and simulated ammonia adsorption on activated carbon with CFD using COMSOL Multiphysics v5.6 (Esposito et al., 2022). In-house experiments were undertaken to validate the intricate modelling of the hydrodynamic and adsorption processes occurring in the system. The Toth adsorption equilibrium isotherm and LDF kinetics were considered, and radial intra-particle mass transfer was attributed to Fickian diffusion. The symmetry in geometry of the ventilator was exploited to reduce domain size and, therefore, computational times.

Qasem and Ben-Mansour developed a Pressure-Vacuum Swing Adsorption (PVSA) framework via CFD to study CO_2 separation from a flue gas mixture using Mg-MOF-74. (Qasem and Ben-Mansour, 2018). Four User-Defined Functions (UDFs) and seven User-Defined Scalars (UDSs) were written to implement the transport equations for the species concentrations, gas temperature, bed temperature and mass transfer to the adsorbent. In line with the work by Augier et al. (2008), constant values for the LDF mass transfer coefficient were taken, neglecting temperature dependencies of considerable consequence (significant in cases of pronounced exothermic adsorption of the heavy product).

Gautier et al. carried out a series of studies (Gautier et al., 2018a,b), the first focusing on developing and validating a 3D CFD Pressure-Swing Adsorption (PSA) model using the commercial software Star-CCM+, and the latter delving into a numerical sensitivity study of a PSA cycle in an isotropic porous medium. Experimental data was well fitted to the Toth isotherm citing thermodynamic consistency, whereas the Langmuir isotherm proved to suffice in the 3-5 bar high-pressure range. Their sensitivity analysis work discusses how the effective thermal conductivity, column dead volume and choice of adsorption equilibrium model influence the robustness of the simulation and the overall cycle performance. The authors acknowledge a shortcoming regarding their approach in how the mass transfer kinetics is modelled using constant mass transfer coefficients, and instead recommend the use of either temperature-dependent LDF mass transfer coefficients or a non-linear driving force (NLDF) model to improve the estimation of local adsorption rates.

The potential of CFD for modelling separation columns is also demonstrated by Subraveti et al. (2018), who formulated a 2D axisymmetric CFD model in ANSYS Fluent v16.2 to study solute propagation in supercritical fluid chromatographic (SFC) columns. The model investigated the effects of viscous fingering and the plug effect on pulse injection. The CFD model could predict key phenomena that are not captured by classical equilibrium-dispersive models, such as a delay in elution times due to inlet pressure increase, peak fronting, and peak distortions. However, it should be noted that the model by Subraveti et al. (2018) studied supercritical fluid chromatography under the isothermal system assumption, leading to a constant linear driving force mass transfer coefficient k. The adsorbing solute was showed a linear equilibrium and the solvent did not adsorb. In contrast, the current work deals with gas adsorption where all the components adsorb, and some (CO_2) in a highly non-linear form. Further, the current model is non-isothermal and the impact of the heat effects on the mass transfer phenomenon (and vice versa) are explicitly considered.

Based on an analysis of published works devoted to the modelling of adsorption processes, it can be concluded that a comprehensive all-in-all validation of 2D and 3D CFD-based fixed bed adsorption models is missing in the literature. It should be noted that 2D and 3D CFD-based studies have a great potential in aiding the design of column internals, choice of adsorbent, and optimization of operating conditions, as well as predicting the process performance at large scales. The applications of a CFD-based adsorption model may include de-risking the scale-up, identifying thermal hot or cold spots, and investigating the impact of flow and heat transfer asymmetries caused by the inlet diffuser design. A relevant scenario for such applications may be found in the field of Direct Air Capture, where the types of contactors are less conventional than those used in more classical separations. Furthermore, the present model may serve as a tool to explore whether certain experimental configurations benefit from in-depth flow resolution with CFD, or can just be modelled using a standard 1D model without concern.

However, it is also observed that the methodologies on most of these research works have certain weaknesses that compromise the accuracy of results or make the model particularly restrictive. A number of studies tend to assume fixed-value parameters, such as heat capacities, conductivities, diffusivities, and –notably– the LDF mass transfer coefficient. Whilst this makes the system easier to solve, it neglects the heavy dependency that this key parameter has on other variables, mainly temperature, diffusivity, and composition. As hinted by Gautier et al., the local adsorption rate may be improved using a LDF model with temperature-dependent mass transfer coefficients (Gautier et al., 2018a). On the other hand, despite the ample capabilities of UDF programming, the literature suggests that it may have not been exploited fully. UDS-based models are available, which try to reproduce transport phenomena equations from scratch; however, a sleeker framework can be obtained by simply adjusting the existing, tried-and-tested equations from commercial CFD solvers using UDFs to include the necessary adsorption terms. Finally, it is observed that literature showcasing a detailed implementation of a CFD-based adsorption model along with its comprehensive validation against experimental data is rare.

Motivated by these facts, our work presents a CFD-based model of the unsteady transport phenomena to reproduce adsorption experiments on a packed column. This work attempts to illustrate in detail the model's implementation into a commercial CFD solver using UDFs written in C. A comprehensive validation of the model is then performed against published experimental data on the single-component and binary competitive adsorption of CO_2 and N_2 on zeolite 13X, which comprises various sets of inlet compositions and operating conditions. Furthermore, the multidimensional capabilities of the model are showcased by exploring the effects of jet inlet injection on the spatial distribution of matter and energy.

It is our intention to offer a thorough description of the construction and validation of our model, which is designed to offer certain advantages over prior studies. Using UDFs in lieu of UDS enables the organic and synergetic integration of the mathematics of adsorption into the well-stablished transport equations from ANSYS Fluent. This favours a cleaner and more straightforward implementation of the model, with significantly less chances of error either in the mathematics of the differential transport equations or in the coding itself. Moreover, our framework allows easy transferability to other adsorbate/carrier combinations including both single-component and multicomponent adsorption on any given adsorbent, meaning that the same model is used and the only changes correspond to the adsorption parameters and experimental conditions. The UDF code has been written with multidimensionality in mind, meaning that the same code is intended to work for both 2D and 3D geometries. This means our model can be applied to various geometries ranging from lab to pilot and even industrial-scale columns. It is equally worth mentioning that the UDF code is modular, and can thus be easily integrated with other custom routines that may be required. Additionally, efforts were made to develop a fully coupled model, where relevant parameters (such as the LDF coefficient and thermodynamic properties) are evaluated locally to strive for accuracy.

Finally, with the purpose of contributing to the collective efforts devoted to adsorption modelling, we are delighted to make this model open-access. The reader is referred to the supporting material, where the Fluent files and UDF code used in this work are free to download, alongside a guided tutorial on how to use them.

2. Computational model

2.1. Scope of the model

The present CFD-based model numerically reproduces the experimental setup and data collected by Wilkins and Rajendran (Wilkins and Rajendran, 2019; Wilkins et al., 2021), where a comprehensive number of Dynamic Column Breakthrough (DCB) experiments on a lab-scale fixed bed were reported for the single-component and binary adsorption of CO_2 and N_2 on zeolite 13X, with He as carrier. A step input of feed containing a certain composition of adsorbate and carrier is sent to the column. As the adsorption ensues, the adsorbing component(s) start to saturate the column gradually in the direction of the flow (from left to right). As a result, material and thermal fronts start developing within the bed. These constitute the basis of a DCB experiment, and are captured by our model and discussed afterwards. For validation purposes, the temperature of the bed at a certain location and the outgoing mole fraction are registered over time, then compared with data from the aforementioned experiments. These results are accompanied by mesh independency and model performance simulations, and are all shown in Section 3.

The model is coded to support single-component and multicomponent/binary adsorption; both are tested and shown in this work. A 2D geometry is used, although the custom UDF code has been written independently of geometrical features or any particular spatial coordinate, meaning that it can also be applied to other 2D and 3D geometries. All validation simulations are carried out using a rectangular axisymmetrical domain. Then, to exploit and explore the advantages of higher-dimensional modelling, a study of gas injection to the same adsorption column is performed on a modified geometry with an inlet jet that more closely resembles the real-life configuration of this system.

2.2. Model description and assumptions

The CFD adsorption model was developed in Fluent v19.2. In concordance with the experimental rig dimensions used by Wilkins & Rajendran, a lab-scale column of 6.4 cm in length and 2.82 cm in internal diameter was designed in ANSYS Meshing. To better reproduce the cylindrical geometry of an adsorption column, a 2D axisymmetric domain was chosen for the present work, although the model can be expanded to higher dimensions. Since the breakthrough experiments are dynamic processes, a time-dependent solution (transient) is enabled. Given that the flow in adsorption processes is typically slow and incompressible, a pressure-based solver was chosen.

The adsorbent bed is modelled using a continuum porous medium model, where the flow is treated as laminar. The velocity within the porous zone is computed using Fluent's physical velocity formulation, which includes the effect of porosity on the transport equation terms and enhances accuracy (Ansys Fluent Theory Guide, 2021). Given that adsorption processes involve a gradual increase in temperature over time, the equilibrium thermal model was used to reproduce the energetics of the system, which assumes that the solid and gas temperatures are the same. As for the fluid phase, the mixing law was used to determine average properties such as the thermal conductivity, specific heat capacity, and viscosity. The individual properties were specified using temperature-dependent polynomial expressions determined from the Fluent database and the CRC Handbook of Chemistry and Physics (CRC Handbook, 2015) (for the thermal conductivity and specific heat capacity), as well as kinetic theory (for viscosity).

A velocity-inlet/outflow-outlet boundary condition set was used, where an artificial outlet extension of the cylinder (henceforth, outflow zone) was installed to ensure convergency and physical results at the column's real exit. See Fig. 1(a) for details. A constant heat transfer coefficient is used, and it is set according to the value reported in the experimental studies (Wilkins and Rajendran, 2019; Wilkins et al., 2021). See further details in the Boundary conditions subsection at 2.5. Finally, second-order discretization schemes were used for all transport equations, except for the species conservation equation, where the third-order MUSCL scheme was applied to enhance the accuracy in material calculations. The SIMPLE algorithm was used to solve the Navier-Stokes equations.

The mathematical model was formulated under the following assumptions:

1. The flow is laminar within the porous zone
2. The gas flow was treated as an incompressible ideal gas following the model described in the work (Tomboulides et al., 1997). The incompressible ideal gas assumption implies that density depends on the operating pressure and local temperature according to the ideal gas law. This assumption is widely used in CFD for modelling gas flows with velocity below the Mach number of 0.3 (Ferziger and Peric, 2002).
3. The fluid phase and adsorbent solid are in thermal equilibrium
4. Buoyancy effects are neglected
5. The influence of the wall channelling effect on the heat and mass transfer near the wall is neglected
6. Radiative heat transfer is neglected
7. The mass transport mechanism is controlled by molecular diffusion in the macropores.

These assumptions determine the form of the governing transport equations, presented next.

2.3. Transport equations

Most CFD environments provide a set of standard transport equations, whose parameters adopt different values according to the specifics of the job. However, when it comes to simulating adsorption processes, CFD software typically lacks actual adsorption models or modules. To overcome this, we have modified the built-in equations by adding source and sink terms to account for the effects that the adsorption process has on each transport phenomena. That is, depending on whether the transport variables are affected positively or negatively as a consequence of adsorption, custom source or sink terms are added accordingly to quantify these effects. This is carried out by coding each additional term via Fluent's User-Defined Functions, which are then hooked to their corresponding transport equation from within the software. The routines were written in C.

With the exception of the momentum conservation equation, all the additional source/sink terms depend on the rate of adsorption of each adsorbed species, $\frac{dq_i}{dt}$. This rate is calculated using the Linear Driving Force (LDF) model, a non-separable differential equation (details on the equation are given in Section 2.4). Hence, solving the transport equations involves solving the LDF model as well, simultaneously. A User-Defined Function was used for this purpose too, as it allowed us to program a fully customizable numerical solver that determines both the rates of adsorption $\frac{dq_i}{dt}$ and the adsorbed quantities q_i at every time step.

Finally, some transport equations –such as the species transport equations– rely on properties that require a mathematical reformulation to account for bed-influenced phenomena, such as dispersion. This is also achieved with UDFs. More details on the programming of our UDFs are provided in Section 2.6.

The modified governing equations are shown and explained below:

- Total mass conservation equation.
  Also known as the continuity equation, this sum typically totals zero, which indicates that the scalar quantity (in this case, mass per unit volume) is neither being created nor destroyed. Here, however, since adsorption is taking place, matter is transferred from the fluid phase to the solid phase. And since the continuity equation governs only the fluid phase, a sink must be created to account for the material 'loss'. Said sink takes the form of the expression highlighted by the underbraces, and is implemented using a UDF macro. The equation is formulated as (Haghpanah et al., 2013; Ansys Fluent Theory Guide, 2021):

$$\varepsilon \frac{\partial \rho_g}{\partial t} + \nabla \cdot \left(\varepsilon \rho_g \vec{v} \right) = \underbrace{-(1-\varepsilon)\rho_p \sum_{i=1}^{n_{comp}} \left(M_i \frac{\partial q_i}{\partial t} \right)}_{UDF} \tag{1}$$

  where ρ_g is the fluid phase (gas mixture) density, $\vec{v}$ is the velocity vector, ρ_p is the adsorbent particle density, ε is the bed void fraction, M_i is the molar mass of the species i adsorbed, and $\partial q_i/\partial t$ is the adsorption rate. The fluid phase density is calculated using Fluent's pressure-based solver and assuming an ideal incompressible gas. The adsorption rate is determined using the LDF equation (see Section 2.4).
- Momentum conservation equation.
  Unlike all other transport equations, the momentum conservation equation does not involve a source term related to adsorption, but rather to the viscous resistance due to the presence of the porous bed. This force term works against the flow, and is proportional to the velocity and inversely proportional to the medium permeability, K. The equation is formulated as (Haghpanah et al., 2013; Ansys Fluent Theory Guide, 2021):

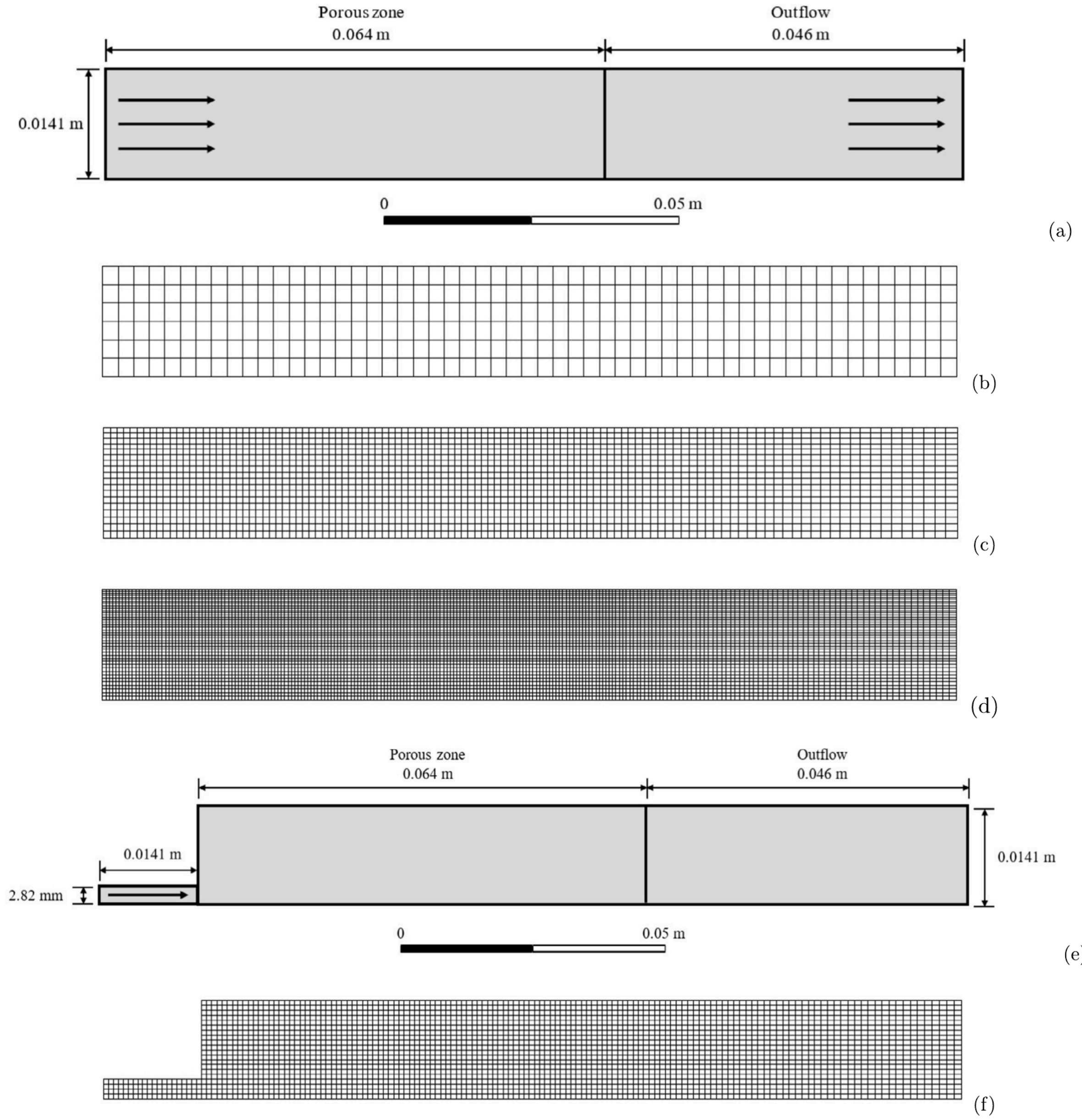

Fig. 1. Diagrams of the 2D computational domains for the adsorption columns studied and their corresponding meshes. (a) Schematic for the flat entrance case, (b) coarse mesh, (c) medium mesh, (d) fine mesh. (e) Schematic for the jet injector case, (f) medium mesh.

$$\varepsilon \frac{\partial\left(\rho_g \vec{v}\right)}{\partial t} + \nabla \cdot \left(\varepsilon \rho_g \vec{v}\vec{v}\right) = -\varepsilon \nabla P + \nabla \cdot \left(\varepsilon \mu_g \left(\nabla \vec{v} + \nabla \vec{v}^T\right)\right) - \frac{\varepsilon^2 \mu_g}{K} \vec{v} \tag{2}$$

where P is static pressure, μ_g is the molecular viscosity, and K is the medium permeability. The second term in the right-hand side corresponds to the stress tensor, whilst the last represents the viscous resistance due to the porous media. The permeability used in this term is calculated using the Carman–Kozeny equation:

$$K = \frac{d_p^2}{150} \frac{\varepsilon^3}{(1-\varepsilon)^2} \tag{3}$$

Since these parameters can be set within the Fluent interface, no UDFs are needed.

- Species mass conservation equation.

The sink term in the species conservation equations follows the same logic as in the continuity equation: it is included to account for the loss of matter in the fluid phase. The difference is that in the mass continuity equation, the adsorbed quantities from all the adsorbing species are summed up, whereas in the species conservation equation for species i, the sink term accounts only for the adsorbed amount of that particular species.

Besides the adsorption sink term, the mixing that takes place as a product of turbulence in the cavities of the packed bed requires a reformulation of the diffusivity. The combination of said mixing and the molecular diffusion D_m can be lumped together as the axial dispersion coefficient, D_L. This term and the aforementioned sink term are formulated through the use of UDFs. The corresponding transport equation is (Haghpanah et al., 2013; Ansys Fluent Theory Guide, 2021):

$$\varepsilon \frac{\partial\left(\rho_g Y_i\right)}{\partial t} + \nabla \cdot \left(\varepsilon \rho_g \vec{v} Y_i\right) = \nabla \cdot \left(\varepsilon \rho_g \underbrace{D_L}_{UDF} \nabla Y_i\right) - \underbrace{(1-\varepsilon)\rho_p \left(M_i \frac{\partial q_i}{\partial t}\right)}_{UDF} \tag{4}$$

where Y_i is the mass fraction of species i, and D_L represents the dispersion coefficient, which is defined as:

$$D_L = 0.7 D_m + v r_p \tag{5}$$

The molecular diffusion, D_m, is calculated locally as a function of the species, temperature and pressure using the Fuller correlation:

$$D_m = \frac{1.013 \times 10^{-2} T^{1.75} \left(1/M_1 + 1/M_2\right)}{P\left((\Sigma_v)_1^{1/3} + (\Sigma_v)_2^{1/3}\right)^2} \tag{6}$$

where Σ_v represent the diffusion volumes of each species. Further details on the programming of these diffusion and dispersion coefficients are provided in Section 2.6.

- Energy conservation equation.
Since adsorption is an exothermic process, the effects of the adsorption heat must be taken into account. This is done through the first source term shown in Equation (7). The released heat is proportional to the adsorption rates of each species, and is calculated based on their corresponding enthalpies of adsorption ΔH_i. These amounts are then summed up to determine the net heat from all adsorbed species.
As for the remaining two sink terms, they are included to account for the thermal mass contribution from the adsorbed matter to the solid bed. Since mass is transferred to the solid phase due to adsorption, the additional adsorbed species increase the overall thermal inertia of the bed according to their individual adsorbed amounts and specific heat capacities; thus, their effect is to lower/buffer the rises in temperature. These two sink terms arise from applying the product rule to the derivative $\frac{\partial}{\partial t}\left((1-\varepsilon)\rho_p q_i M_i c_{p,a} T\right)$ –which is originally inside the transient term–, and are then switched to the right-hand-side with negative signs (thus becoming sink terms). With all these considerations in mind, the modified energy conservation equation is expressed as (Haghpanah et al., 2013):

$$\begin{aligned}
&\frac{\partial}{\partial t}\left(\varepsilon \rho_g c_{p,g} T + (1-\varepsilon)\rho_p c_{p,p} T\right) + \nabla \cdot \left(\varepsilon \rho_g \vec{v} c_{p,g} T\right) \\
&\quad = \nabla \cdot \left(\varepsilon \kappa_{eff} \nabla T\right) \\
&\quad \underbrace{+ (1-\varepsilon)\rho_p \sum_{i=1}^{n_{comp}} \left((-\Delta H_i)\frac{\partial q_i}{\partial t}\right)}_{UDF} \underbrace{- (1-\varepsilon)\rho_p T \sum_{i=1}^{n_{comp}} \left(c_{p,a_i} M_i \frac{\partial q_i}{\partial t}\right)}_{UDF} \\
&\quad \underbrace{- (1-\varepsilon)\rho_p q_i \sum_{i=1}^{n_{comp}} \left(c_{p,a_i} M_i\right) \frac{\partial T}{\partial t}}_{UDF}
\end{aligned} \tag{7}$$

where $c_{p,g}$ is the gas mixture specific heat capacity, $c_{p,p}$ is the adsorbent particle specific heat capacity, c_{p,a_i} is the adsorbed phase specific heat capacity of species i, κ_{eff} is the effective thermal conductivity, $(-\Delta H_i)$ is the heat of adsorption of species i, and q_i is the solid phase loading of species i.
The effective thermal conductivity is calculated as a voidage-weighted average between the gas-phase and solid-phase thermal conductivities, as in Equation (8):

$$\kappa_{eff} = \varepsilon \kappa_g + (1-\varepsilon)\kappa_p \tag{8}$$

The specific heat capacity of the adsorbed phase for each adsorbate was determined with respect to their corresponding gas phase c_p values according to the following rule, as explored in works Rahman et al. (2013); Al-Muhtaseb and Ritter (1999); Walton and LeVan (2005); Men'shchikov et al. (2020):

$$c_{p,a} = \left(\frac{\partial h_g}{\partial T}\right)_P - \left(\frac{\partial Q_{iso}}{\partial T}\right)_P = c_{p,g} - \left(\frac{\partial Q_{iso}}{\partial T}\right)_P \tag{9}$$

Here, the isosteric heat, $Q_{iso} = -\Delta H$ has a linear dependency on the temperature according to $-\Delta H = -\Delta U + RT$. The values of the internal energy for each species are given in Table 5. The isosteric heat of CO_2 on zeolite 13X, Q_{iso,CO_2} [J/mol], is also significantly dependent on loading, according to the following empirical relationship determined by Wilkins and Rajendran (2019):

$$Q_{iso,CO_2} = -5156\, q_{CO_2} + 50907 \tag{10}$$

2.4. Adsorption rate modelling: linear driving force (LDF) model

As introduced previously, this differential equation is used to determine the rate of adsorption. It is formulated as a mass transfer equation for the solid phase, where the rate of change is proportional to the difference between the equilibrium and current solid phase loadings (Wilkins and Rajendran, 2019):

$$\frac{\partial q_i}{\partial t} = k_i\left(q_i^* - q_i\right) \tag{11}$$

where k_i is the LDF mass transfer coefficient, q_i^* is the equilibrium solid loading, and q_i is the actual solid loading of species i. Assuming that the transport mechanism is controlled by molecular diffusion within the macropores (which is the case for the adsorption of CO_2 and N_2 on zeolite 13X), the adsorption mass transfer coefficient, k_i, is formulated as the following lumped parameter (Wilkins and Rajendran, 2019):

$$k_i = \frac{c_i}{q_i^*}\frac{15\varepsilon_p D_{eff}}{r_p^2} \tag{12}$$

where ε_p is the particle porosity and D_{eff} is the effective diffusion. The latter, D_{eff}, is expressed as a function of the molecular diffusion, D_m, according to:

$$D_{eff} = \frac{D_m}{\tau} \tag{13}$$

The molecular diffusivity of the gas phase, D_m, is calculated using the Fuller correlation as described in Equation (6).

There are as many LDF equations as there are adsorbing species in a system. Our UDF framework has been setup for two adsorbing species, but more can be added as needed. The LDF equation solves for the solid phase loading, q_i, for all the adsorbing species i. The equilibrium solid phase loading, q_i^*, is calculated from an appropriate adsorption isotherm $q_i^* = f(c_i, P, T)$ (Haghpanah et al., 2013; Wilkins and Rajendran, 2019). The form of this function depends not only on the equilibrium data for a particular system, but also on the isotherm model and fitting procedure employed, all of which involve different parameters and provide varied predictions. In that sense, several isotherm formulations arise, such as Single and Dual Site Langmuir (SSL and DSL). For this particular case, Dual Site Langmuir isotherms are used to reproduce the adsorption equilibrium of CO_2 and N_2 on zeolite 13X, since they are reported to produce a good predictive description for both single-component and competitive breakthrough experiments in the reference paper Wilkins and Rajendran (2019). Moreover, the parameters of the DSL model are determined by fitting procedures with different constraints and underlying assumptions regarding the solid loadings. The procedures chosen will be indicated for each case in Section 3. The DSL model is represented by the following equation:

$$q_i^* = \frac{q_{sb,i} b_i c_i}{1 + \sum_{i=1}^{n_{comp}} b_i c_i} + \frac{q_{sd,i} d_i c_i}{1 + \sum_{i=1}^{n_{comp}} d_i c_i} \tag{14}$$

where q_{sb} and q_{sd} are the saturated solid phase loadings of adsorption sites 1 and 2 respectively, and c_i is the molar concentration of species i. The coefficients b_i and d_i are determined using the following Arrhenius-type expressions (Haghpanah et al., 2013):

$$b_i = b_{0,i} \exp\left(\frac{-\Delta U_{b,i}}{RT}\right) \tag{15}$$

$$d_i = d_{0,i} \exp\left(\frac{-\Delta U_{d,i}}{RT}\right) \tag{16}$$

The molar concentration c_i is determined from the mass fractions Y_i (passed from the species mass balances) using the following relationship:

$$c_i = \frac{P}{RT} \frac{Y_i/M_i}{\sum_{j=1}^{n_{comp}} \frac{Y_j}{M_j}} \tag{17}$$

The isotherm parameters used in this work were determined experimentally by Wilkins & Rajendran, and are used here for adsorption equilibrium modelling. They are shown in Table 5. The global parameters of the model are shown in Table 6.

2.5. Boundary conditions

The boundary condition for the momentum conservation equation at the inlet is defined as the superficial feed velocity, whilst the outlet boundary condition is of the outflow type (Haghpanah et al., 2013):

$$v\Big|_{z=0} = v_{feed,sup} \tag{18}$$

$$\frac{dv}{dz}\Big|_{z=L} = 0 \tag{19}$$

Referring to the theory of fluid dynamics, momentum conservation and transport equations for the porous media can be written using two formulations: the superficial velocity porous formulation and the physical velocity formulation (Ansys Fluent Theory Guide, 2021). It is worth mentioning that the physical velocity formulation is considered across the porous medium. When this formulation is used, the velocity values in the porous region increase automatically according to $\vec{v}_{phys} = \vec{v}_{sup}/\varepsilon$, which allows us to determine the true or physical velocity across this region. Not doing so would limit the accuracy of the porous media approach, as the velocity increase due to the packing would be neglected. This automatic correction of velocity is particularly convenient when simulating columns composed by both porous regions (packed beds) and free-flow regions (e.g. inlet/outlet pipes), such as the inlet injection system shown in Section 4.

In the case of the species and energy conservation equations, since we intend to model adsorption across a packed bed where dispersion has a significant influence, two sets of Robin boundary conditions are used at the inlet (Haghpanah et al., 2013):

$$\frac{dY_i}{dz}\Big|_{z=0} = -\frac{1}{D_L} v_{sup}\Big|_{z=0} \left(Y_{i,feed} - Y_i\Big|_{z=0}\right) \tag{20}$$

$$\frac{dT}{dz}\Big|_{z=0} = -\frac{\rho c_p}{k_{eff}} v_{sup}\Big|_{z=0} \left(T_{feed} - T\Big|_{z=0}\right) \tag{21}$$

In the modelling of dispersed plug flows, these are also known as Danckwerts' boundary conditions, and account for the immediate influence of dispersion when the flow enters the bed. Both boundary conditions above can be implemented in Fluent by enabling Inlet Diffusion (Ansys Fluent Theory Guide, 2021). As for the outlet, outflow boundary conditions are applied (Haghpanah et al., 2013):

$$\frac{dY_i}{dz}\Big|_{z=L} = 0 \tag{22}$$

$$\frac{dT}{dz}\Big|_{z=L} = 0 \tag{23}$$

At $r = R_W$, the no-slip condition is used for the momentum equation. Furthermore, a convective boundary condition is set up to account for external heat transfer, and zero flux is defined for the species boundary condition (Haghpanah et al., 2013):

$$\frac{dY_i}{dr}\Big|_{r=R} = 0 \tag{24}$$

$$-k\frac{dT}{dr}\Big|_{r=R} = h\left(T_w - T_{inf}\right) \tag{25}$$

At $r = 0$, symmetry boundary conditions are used for all the transport equations.

As for the LDF differential equation, since it doesn't involve derivatives with respect to space, no boundary conditions apply, only initial values.

2.6. User-defined functions (UDFs)

The custom mathematical reformulations discussed in Section 2.3 are integrated into the CFD software using UDFs. So far, the term 'UDF' has been used to convey the idea of custom code. Specifically, a UDF is a C program that is comprised of several sub-routines called macros, in which the actual custom code is implemented. There are different types of macros, each suited to particular customization goals. In the present job, one fully fledged UDF file was written, comprised of ∼550 lines and the following 9 macros:

- `DEFINE_ADJUST` macro (1 instance)
  The purpose of this macro is to execute instructions at the start of every iteration. It was thus used to implement a numerical scheme to solve the Linear Driving Force (LDF) differential equation. In this way, the solid phase concentration q_i and rate of adsorption $\frac{dq_i}{dt}$ are calculated first, and can then be passed to the subsequent transport equations.
  With regard to the numerical scheme, two different approaches were followed during the development of this work. The first consisted in an explicit method. This scheme is based on the 4th-order Runge-Kutta numerical method, which made it possible to determine the value of q_i at the current time step using known quantities from the previous time step. The advantage of this method is that the LDF equation can be solved straightforwardly, at the very first iteration in each time step, which results in few overall iterations per time step. However, its main drawback lies in the computational time. Since explicit methods are conditionally stable, very small time steps had to be chosen to guarantee accuracy and stability in the solution (such as $dt = 0.001$). The discretization of the LDF equation using this method is shown below, for completeness:

$$k_{RK,1} = h(k_i(q_i^* - q_i^t)) \tag{26}$$

$$k_{RK,2} = h(k_i(q_i^* - (q_i^t + k_{RK,1}/2))) \tag{27}$$

$$k_{RK,3} = h(k_i(q_i^* - (q_i^t + k_{RK,2}/2))) \tag{28}$$

$$k_{RK,4} = h(k_i(q_i^* - (q_i^t + k_{RK,3}))) \tag{29}$$

$$q_i^{t+\Delta t} = q_i^t + \frac{1}{6}(k_{RK,1} + 2k_{RK,2} + 2k_{RK,3} + k_{RK,4}) \tag{30}$$

  The second and definite approach was an implicit formulation. This was made possible using the trapezoidal method (second order). This formulation requires that the LDF equation be solved simultaneously alongside the transport equations, on every iteration. Since this yields unrefined values of q_i and $\frac{dq_i}{dt}$ during the first iterations, more iterations need to be allocated to reach sufficiently low residuals. However, this is greatly compensated by the fact that the implicit scheme allows time steps to be 10 to 30 times larger than the explicit scheme, which makes simulating a full DCB experiment feasible within a reasonable time frame. The mathematical implementation for the second-order implicit scheme is shown below:

$$q_i^{n+1} = \frac{q_i^t + \frac{dt}{2}\left(k_i^t(q_i^{*,t} - q_i^t) + k_i^{n+1} q_i^{*,n+1}\right)}{1 + \frac{dt}{2}k_i^{n+1}} \tag{31}$$

  Here, n is the iteration number. The rate of adsorption is discretized as:

$$\frac{dq_i}{dt} = \frac{q_i^{n+1} - q_i^t}{dt} \tag{32}$$

To enhance the stability of the LDF implicit formulation even further, relaxation factors are also implemented in line with Equation (33). Given the stiff nature of the LDF equation (especially with high values of k_i), low relaxation factors significantly enhance stability, albeit at the cost of an extra number of iterations.

$$q_i^{n+1} := RLX q_i^{n+1} + (1 - RLX) q_i^n \tag{33}$$

After a series of tests, it was determined that a relaxation factor of $RLX = 15$ provided a good balance between stability and speed.

- `DEFINE_INIT` macro (1 instance)
  The name of this macro stands for 'initialization', and as its name implies, it is used to execute instructions at the very start of a job. This macro was thus mainly used to specify initial values for the LDF variables (namely, q_i), as these are excluded from the standard initialization routine in Fluent since our custom LDF model is not a standard Fluent equation. Given that the breakthrough experiments are started with a clean and desorbed column, q_i is initialized as zero.
- `DEFINE_DIFFUSIVITY` macro (1 instance)
  The purpose of this macro is to re-define the mass diffusivity of the fluid phase at every cell in the domain. It that regard, it is suited for the localized calculation of the molecular diffusion in line with Equation (6), and the subsequent calculation of the dispersion coefficient in line with Equation (5). The resulting value is then passed to the species conservation equations.
- `DEFINE_EXECUTE_AT_END` macro (1 instance)
  As its name implies, this macro executes instructions at the end of a given time step. It was thus used to save the converged values of q_i^t in a User-Defined Memory, so that it can later be used in the iterations from the next time step.
- `DEFINE_SOURCE` macro (5 instances)
  This macro allows source or sink terms to be directly added to any of the standard Fluent transport equations. It was thus used to code and add each of the source/sink terms discussed in Section 2.3. As opposed to the previous macros, five instances of the source macro were needed: one for the continuity equation sink, two for the species equation sinks (one for each adsorbent species), and two for the energy equation source and sink terms (one instance for the source term, and another instance for the two sink terms).

To further illustrate the usage of UDF macros in our model, the reader is referred to Table 7, where a summary description of the purpose and output of each macro is given, alongside the relevant equations that were coded in each.

3. Model validation

Upon the complete implementation of the CFD model plus UDFs, we test its validity against published experimental results by Wilkins and Rajendran (2019). In their study, several single and multicomponent Dynamic Column Breakthrough (DCB) experiments were performed. The adsorbent of choice was zeolite 13X. As mentioned before, these experiments were carried out on a lab-scale column of 0.064 m in length and 0.0282 m in diameter. Hence, due to the cylindrical configuration of the system, it was set up as a 2D axisymmetric case in Fluent. An schematic representation of the column's domain in Fluent is shown in Fig. 1(a). As briefly stated in Section 2.2, the 'outflow' zone was added with the purpose of ensuring that the influence of the outflow boundary conditions was not carried back to the actual column exit, so that accuracy and stability can be guaranteed.

In concordance with the experimental procedures, all our simulations start with the column being fully cleansed or unsaturated ($q_{N_2} = q_{CO_2} = 0$) and filled completely with helium, which is assumed to be non-adsorbing and used as carrier gas. Then, at $t = 0$, a gas mixture with various compositions is fed into the column. As the adsorbing species in the mixture becomes adsorbed, the column starts to saturate gradually with this component from left to right. As a consequence, inspecting the 2D contour of a given variable like temperature or adsorbate mole fraction reveals an advancing red section through the column. This is known as a front, and when it reaches the end of the column, a breakthrough is said to have occurred. This is, in short, a Dynamic Column Breakthrough (DCB) experiment: it serves as a benchmark to test the validity of adsorption models, and it forms the basis of the simulations performed in this work.

Before the complete set of validation simulations are presented, results from a mesh independency study are first discussed alongside a performance analysis of our computational model. These preceding steps were crucial for determining a proper mesh resolution, time step, and number of iterations for the rest of the simulations. To this end, one experimental case was chosen from Wilkins' work (Wilkins and Rajendran, 2019), involving 351 ccm of a mixture of 50/50% mole N_2/He entering the column at 0.96 bar and 23 °C. These results are explored next (Sections 3.1 and 3.2). The model validation results are presented from Section 3.3 onwards.

3.1. Mesh independency study

In order to guarantee mesh independency, three meshes were produced, each with a different level of refinement. For quick reference, they have been labelled coarse, medium, and fine. These meshes are shown in Fig. 1(b,c,d), and the details on their sizes can be seen in Table 1.

To estimate an appropriate value for the simulation time step for this study, the CFL number (Courant – Friedrichs – Lewy) was used. According to the CFL number formulation, the maximum time step, Δt^{max} is calculated according to (34):

$$\Delta t^{max} = \frac{\Delta z}{v} \tag{34}$$

where v is the flow velocity and Δz is the grid size in the z direction. The results of this calculation are shown in Table 2. From this table, we observe that the minimum value for Δt^{max} is around 0.02 s. However, since an implicit formulation is used –both by the Fluent solver and by our custom LDF solver–, the stability limit cap can be increased. As seen later in Performance Study (Section 3.2), a value of $dt = 0.03\,s$ provides a very accurate result, comparable to that of a very fine time step. Hence, the mesh independency study for all three cases (coarse, medium and fine) is performed using a time step of 0.03 seconds for uniformity.

The results of the mesh independency study are shown in Fig. 2. They correspond to the molar composition breakthrough plot and temperature breakthrough plot. In this figure and all subsequent ones, the simulated breakthrough composition was determined by an area average at the column outlet plotted over time, whilst the temperature results were registered both at the edge and at the centre of the column at $z = 0.8L$, in accordance with the experiment measurements. As can be seen in Fig. 2, the model is capable of reproducing the composition breakthrough with a good level of accuracy and independently of the mesh used, crossing through most of the experimental data points. As for the temperature breakthrough, even though the model's curve does not exactly match all the data points, the breakthrough peak is resembled closely, and the same downward tendency is reproduced with an overall deviation of roughly 0.5 °C, which can be deemed acceptable considering some degree of error in the temperature measurements.

It is worth mentioning that the results in Fig. 2 are typical in unidimensional adsorption modelling. However, the main benefit of performing a higher-dimensional simulation lies in the enhanced spatial insight. This is demonstrated, for instance, in field contours and radial plots, such as the ones shown in Fig. 3. The inlet composition was switched to 85/15 N_2/He to better appreciate the temperature gradients, since this composition allows a higher temperature rise. In Fig. 3(a), two snapshots of the temperature contour are shown at two different times:

Table 1
Characteristics and cell count for the three meshes used in the grid resolution study.

Mesh	Axial cell divisions		Radial cell	Number of	Total number
	Main (porous zone)	Outflow	divisions	cells (Main)	of cells
Coarse	32	23	6	192	330
Medium	75	40	18	1350	2070
Fine	150	75	40	6000	9000

Table 2
CFL numbers for the three meshes used in the grid resolution study.

Mesh	Interstitial Velocity [m/s]	Main section Length [m]	Δx [mm]	Δt^{max} [s]
Coarse	0.02335	0.064	2.0000	0.0857
Medium	0.02335	0.064	0.8533	0.0365
Fine	0.02335	0.064	0.4267	0.0183

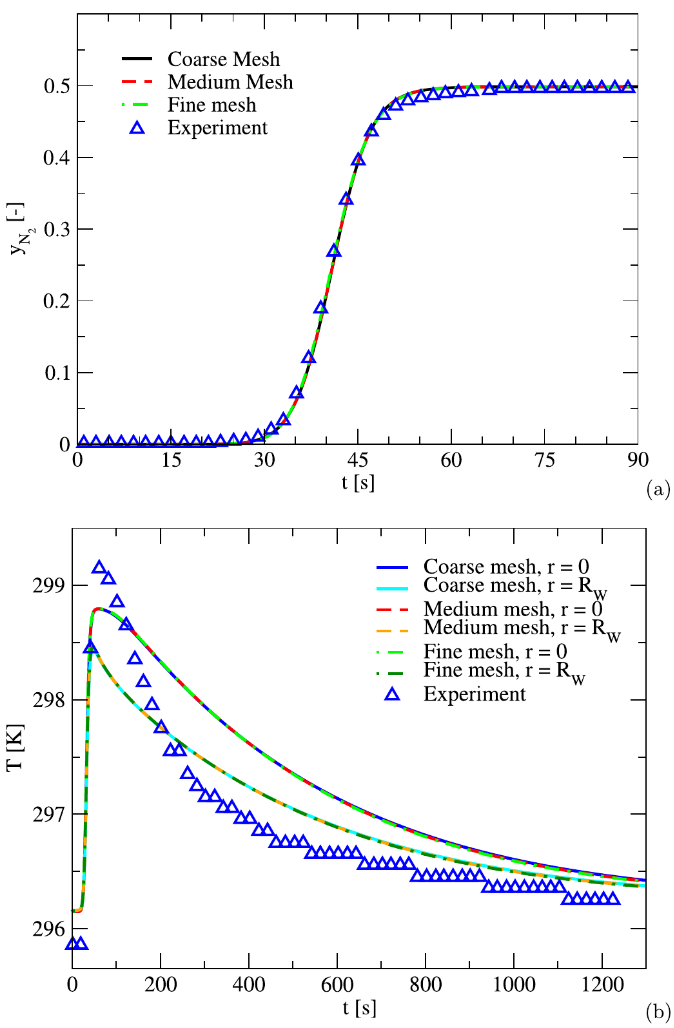

Fig. 2. Mesh independency results for the 50/50% mole N_2/He case. (a) Composition breakthrough plot. (b) Temperature breakthrough plot. Here, 'Experiment' refers to the work by Wilkins (Wilkins and Rajendran, 2019).

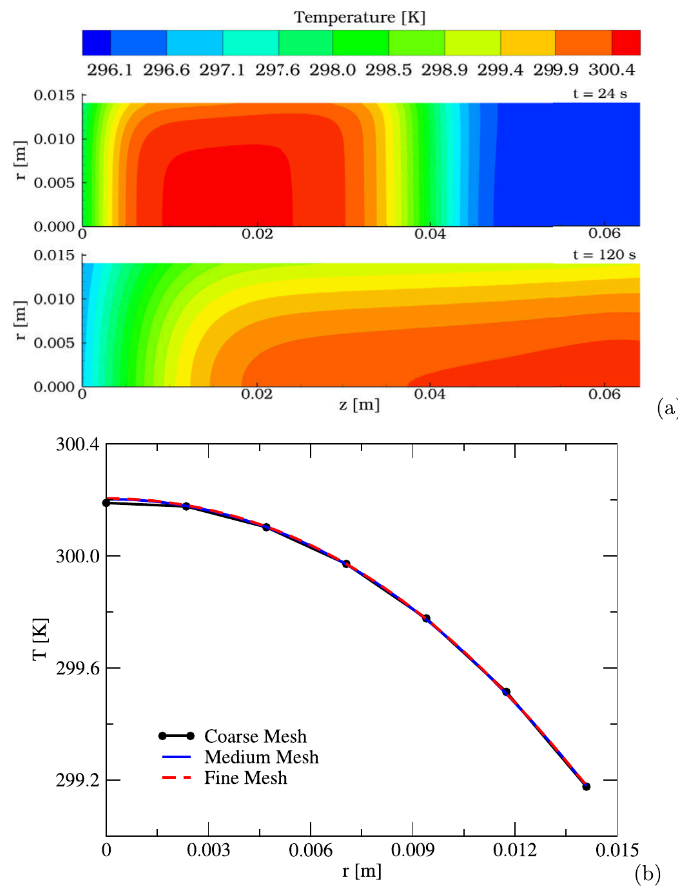

Fig. 3. 2D distribution of temperature for an inlet composition of N_2/He 85/15%. (a) Temperature contours at $t = 24\,s$ (top, before breakthrough) and $t = 120\,s$ (bottom, after breakthrough), fine mesh. (b) Radial temperature profile across the column at $z = 0.0512\,m$ (midline) and $t = 120s$ for the different mesh sizes.

$t = 24s$ and $t = 120s$. In the adsorption time reference, these two times correspond to when the hot front (due to the release of adsorption heat) is travelling around the middle of the column (24 s), and many seconds after breakthrough has occurred and the bed is gradually cooling down (120 s). As can be seen in both contour plots, temperature gradients are present within the bed, both in z and R, especially during the cooling stage. The thermal gradient in R can be better appreciated by means of a radial plot, as demonstrated in Fig. 3(b) at $t = 120s$. The decay in the temperature towards the wall is evident and consistent with what is expected due to heat transfer to the surroundings. Furthermore, the radial plots also allow us to compare the variance in the results due to the mesh resolution. As can be seen, the results from the medium and coarse meshes overlap precisely. A more rugged curve is obtained for the coarse mesh, due to the lower number of grid points. Nevertheless, despite the low grid-point density of the coarse case, its values are very close to those of the medium and fine meshes, with a maximum difference of only 0.016 K from the fine mesh, which showcases the robustness and stability of the present CFD model even at low grid resolution.

Besides the quality of the simulation results, another important factor to consider is the computational performance of each mesh resolution. Details on the speed in each case are provided in Table 3. As can be seen, in relation to the fine mesh speed, the simulations with the medium mesh can be up to 2.3 times faster. The coarse mesh, in turn, is 4 times faster. Given that the medium mesh provides virtually the same

Table 3
Performance results for each mesh.

Mesh	Total number of cells	Amount of DCB data simulated, in seconds [s_{DCB}]	Elapsed time, real life [h]	Computation Speed [s_{DCB}/h]
Coarse	275	1500	3.78	396.5
Medium	2070	1500	6.63	226.1
Fine	9000	1500	15.48	96.9

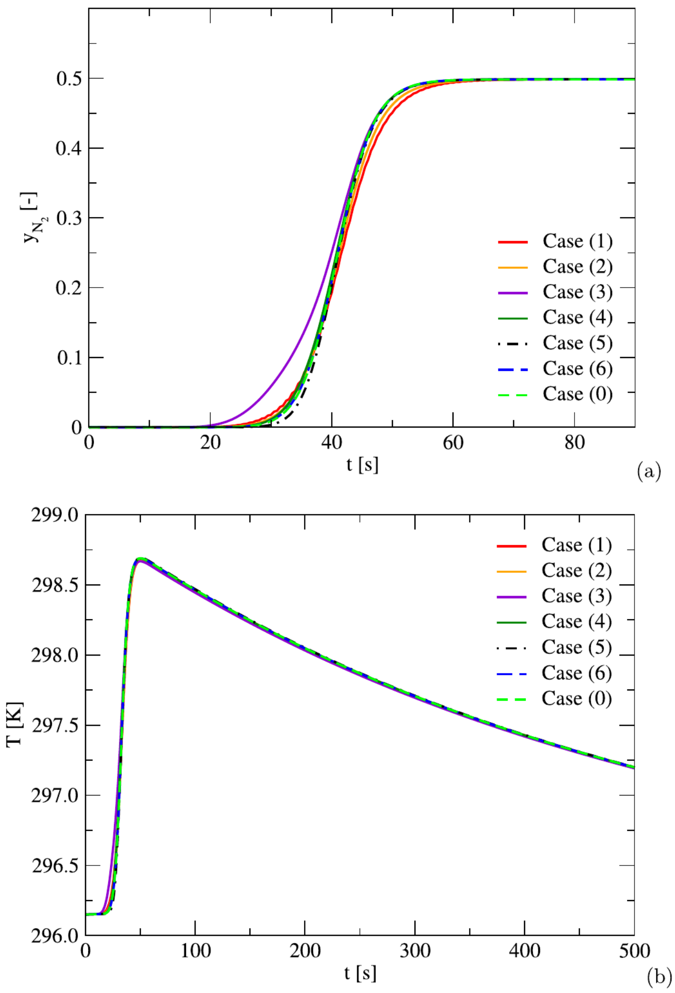

Fig. 4. Results of the time step and iteration number study on the 50/50% mole N_2/He case. (a) Molar fraction breakthrough plot. (b) Radial average temperature breakthrough plot ($z = 0.8\,L$). Time step [s] and iterations: (1) 0.5, 50; (2) 0.5, 125; (3) 0.1, 25; (4) 0.1, 50; (5) 0.03, 25; (6) 0.03, 50; (0, reference) 0.001, 50.

accuracy as the fine mesh at a significantly higher speed, it can thus be deemed the best performing mesh.

3.2. Performance study

Next, in order to gain further insights into how computational speed correlates to accuracy, a performance study was carried out focusing on time step and iterations. For this study, various combinations of time step and maximum iterations-per-time-step (henceforth, simply 'iterations') were considered, resulting in 6 cases. An additional Case (0) with 50 iterations and a time step of $t = 0.001\,s$ was also simulated, and is considered our reference for 'true' accuracy given its very fine time step and relatively large number of iterations. The medium mesh resolution was used. Each case and its relevant computational statistics are described in Table 4, where the Root Mean Square Error (RMSE) was calculated for each case with respect to the reference case as a means of quantifying their accuracy. The composition and breakthrough curves for all cases are shown in Fig. 4. These computations, as well as all the simulations presented in this work, were performed on an openSUSE Leap 15.4 Linux computer using a single 13th Gen Intel® Core ™ i9-13900KS processor (4.3 GHz) in serial mode.

As can be observed from Fig. 4, different combinations of time step and iterations result in slightly shifted temperature and (especially) composition breakthrough curves. It can be seen that the accuracy of the 6 cases depends on the specific combination of iterations and time steps. For instance, Case (6), which has the finest time step compared to cases 1-4 (the reference case, 0, is not counted), is the most accurate of the six, as evidenced by it having the lowest RMSE and by how it overlaps Case (0) in the plots precisely. This may be attributed to the fact that it not only has an adequate time step, but is also allocated a relatively large number of iterations: 50. However, if fewer iterations are provided, then the accuracy is affected. Let us take, for instance, Case (5) as an example. Despite having the same time step as the most accurate Case (6), Case (5) displays a lower accuracy (higher RMSE), even lower than Case (4), which has a larger time step. This is explained by the fact that fewer iterations were allocated for Case (5): 25, whilst Case (4) operates under 50 iterations, which not only compensate the larger time step, but makes this case twice as fast as Case (5). The accuracy of Case (3) –similarly to Case (5)– pales in comparison to its counterpart (Case 4, same time step), which is again due to the lower number of iterations. Finally, the large time step cases, (1) and (2), are the least accurate of all. However, it is worth mentioning that by allocating up to 125 iterations, Case (2) achieves an accuracy comparable with that of Case (5) (similar RMSE), whilst being more than 7 times faster.

The importance of the number of iterations can be explained by the use of a low relaxation value for solving the LDF equation: even though it increases the stability of the solver, it requires more iterations per time step to achieve a low enough residual.

In summary, this study allows us to appreciate that a good balance between computational speed and accuracy can be achieved by choosing a sufficiently low time step whilst allowing a sufficiently large numbers of iterations. If a larger time step is chosen, it must be accompanied by an increase in the number of iterations as well.

Case (6), with $t = 0.03\,s$ and 50 iterations per time step, provided the highest accuracy, which is confirmed both by its minimal RMSE and by how it visually overlaps Case (0). Hence, these two values for the iterations and time step were chosen for all of the simulations that follow in this work. Finally, it must be acknowledged that, despite the slight variations, almost all cases (excepting Case 3) are still in close agreement with one another, especially regarding the temperature breakthrough, which gives us the confidence of –if needed– increasing the time step for very long simulations for the sake of speed, whilst still expecting reasonably good results.

3.3. Single-component adsorption of N_2

Based on the learnings from the mesh independency and performance study, it was decided that a good combination of accuracy and computational cost is provided by the medium mesh and a time step of 0.03 s, with up to 50 iterations to aid convergence. With these characteristics, three different experiments were reproduced for the following inlet molar compositions of N_2/He: 85/15%, 50/50% and 25/75% mole. The flow rates are 350, 351, and 350 ccm respectively, and the

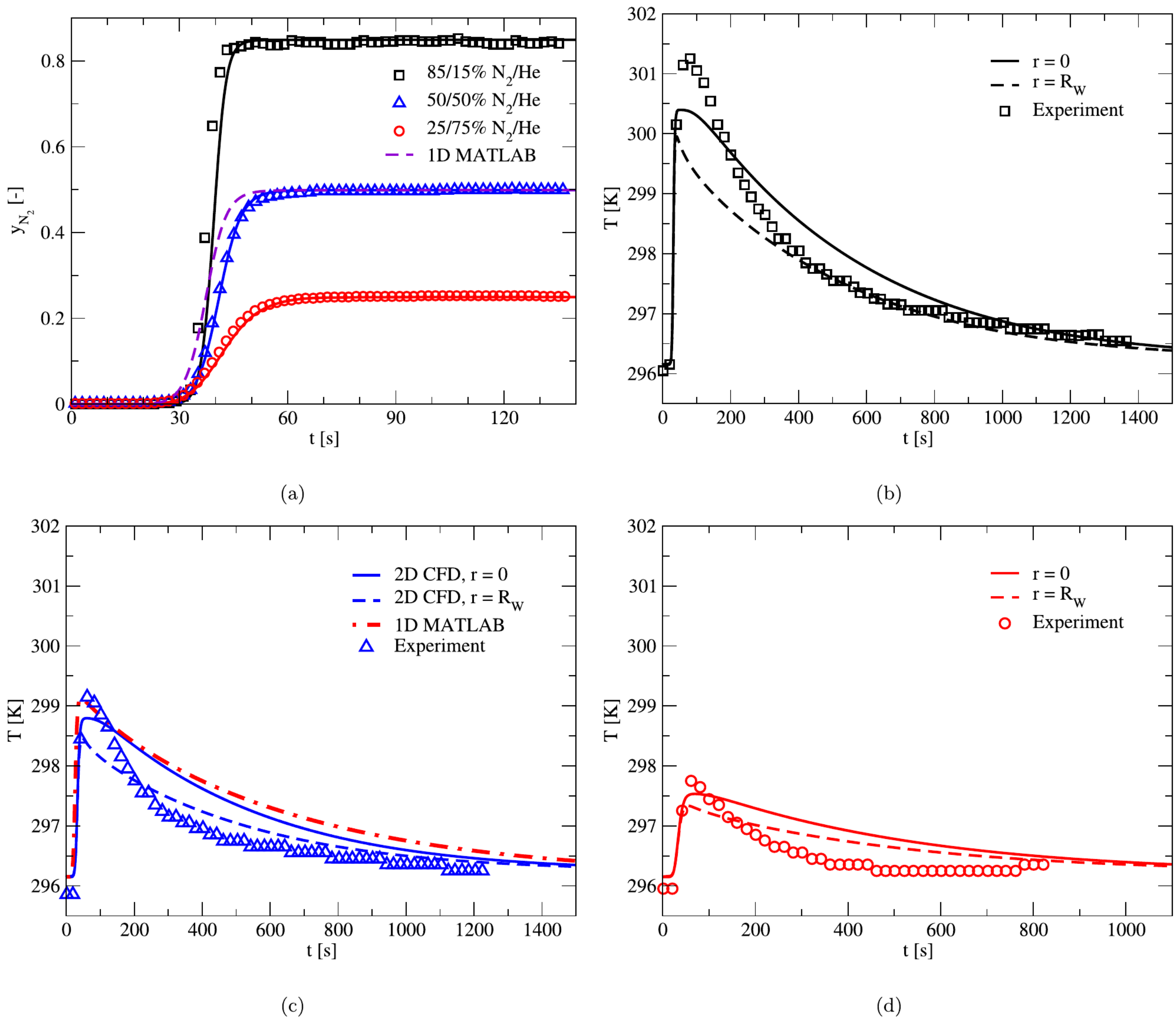

Fig. 5. N_2 composition and temperature breakthrough curves predicted numerically for different N_2/He inflow gas compositions. Here, the lines represent the CFD results and the symbols refer to experimental data by Wilkins (Wilkins and Rajendran, 2019), 1D MATLAB refers to data by Wilkins (Wilkins and Rajendran, 2019). (a) Composition breakthrough of all cases. (b) Temperature breakthrough, 85/15% mole N_2/He case. (c) Temperature breakthrough, 50/50% mole N_2/He case. (d) Temperature breakthrough, 25/75% mole N_2/He case.

Table 4
Performance results for each case at different time steps and iterations.

Case	dt [s]	Iterations per time step	Amount of DCB data simulated, in seconds [s_{DCB}]	Elapsed time, real life [min]	Computation Speed [s_{DCB}/h]	RMSE
(1)	0.5	50	500	16	1875.0	1.161×10^{-2}
(2)	0.5	125	500	17	1764.7	6.038×10^{-3}
(3)	0.1	25	500	37	810.8	2.440×10^{-2}
(4)	0.1	50	500	58	517.2	3.533×10^{-3}
(5)	0.03	25	500	125	240.0	4.319×10^{-3}
(6)	0.03	50	500	158	189.9	4.074×10^{-4}
(0)	0.001	50	500	2801	10.7	Reference

Table 5
Isotherm parameters used for adsorption equilibrium modelling, for both CO_2 and N_2.

Gas	i	Model	$q_{b,i}^{sat}$ [mol/kg]	b_0 [m^3/mol]	ΔU_b [J/mol]	$q_{d,i}^{sat}$ [mol/kg]	d_0 [m^3/mol]	ΔU_d [J/mol]
CO_2	1	DSL	3.257	2.09×10^{-7}	-42670	3.240	1.06×10^{-7}	-32210
N_2	2	EES	3.257	2.13×10^{-6}	-16250	3.240	2.13×10^{-6}	-16250
N_2	2	UES	3.257	7.96×10^{-7}	-18860	3.240	6.94×10^{-7}	-17760

Table 6
Parameters used for CFD-based adsorption modelling.

Parameter	Value / Method	Unit
Column characteristics		
Column length, L	0.064	m
Column inner radius, R_{in}	0.0141	m
Column outer radius, R_{out}	0.0159	m
Artificial outflow zone length	0.046	m
Mean particle diameter, d_p	0.001	m
Bed porosity, ε	0.4	Dimensionless
Particle porosity, ε_p	0.35	Dimensionless
Particle tortuosity, τ	3	Dimensionless
Properties		
Column wall density, ρ_w	8030	kg/m^3
Column wall specific heat capacity, $c_{p,w}$	502.48	J/kg·K
Column wall thermal conductivity, κ_w	16.27	W/m·K
Adsorbent particle density, ρ_p	1050	kg/m^3
Adsorbent particle specific heat capacity, $c_{p,p}$	856	J/kg·K
Adsorbent particle thermal conductivity, κ_p	0.3	W/m·K
Fluid phase (gas mixture) density, ρ_g	incompressible ideal gas	kg/m^3
Fluid phase specific heat capacity, $c_{p,g}$	mass-weighted mixing law	J/kg·K
Fluid phase thermal conductivity, κ_g	mass-weighted mixing law	W/m·K
Fluid phase viscosity, μ_g	mass-weighted mixing law	Pa·s
Pure gas specific heat capacity	polynomial function of T	J/kg·K
Pure gas thermal conductivity	polynomial function of T	W/m·K
Pure gas viscosity	kinetic theory	Pa·s
Adsorbed phase specific heat capacity, $c_{p,a}$	Equation (9)	J/kg·K
Effective thermal conductivity, κ_{eff}	Equation (8)	W/m·K
Heat transfer coefficient to exterior, h_{out}	10	$W/m^2 \cdot K$
Molecular diffusivity, D_m	Fuller correlation	m^2/s
Dispersion coefficient, D_L	Equation (5)	m^2/s
Constants		
Universal gas constant, R	8.3144626	J/mol·K
Molar mass of CO_2, M_{CO_2}	0.04400995	kg/mol
Molar mass of N_2, M_{N_2}	0.0280134	kg/mol
Molar mass of He, M_{He}	0.004002602	kg/mol
Diffusion volume of CO_2, $(\Sigma_v)_{CO_2}$	26.9	-
Diffusion volume of N_2, $(\Sigma_v)_{N_2}$	17.9	-
Diffusion volume of He, $(\Sigma_v)_{He}$	2.88	-

Table 7
Description of all UDF macros employed in the CFD adsorption model.

UDF Macro	Main purpose	Output	Equation(s)
DEFINE_INIT	Used for initializing the values of the scalar q_i from the LDF equation.	The initial value of q_i is set to zero at the beginning of the simulation (no adsorbed species).	(11)
DEFINE_ADJUST	Used for specifying isotherm parameters & auxiliary adsorption equations, and for implementing a numerical solver for the LDF equation.	The values of c_i, q_i, q_i^*, k_i and dq_i/dt are calculated and stored in User-Defined Memories (UDMIs), ready to be passed to the Fluent's solver where needed.	(11), (12), (13), (14), (15), (16), (17), (31), (32), (33)
DEFINE_EXECUTE_AT_END	Used for storing the definitive values of q_i and dq_i/dt at the end of each time step.	The values of q_i and dq_i/dt are stored in UDMIs to be used in the next set of iterations within the numerical LDF solver.	(11)
DEFINE_SOURCE (Continuity)	Used for implementing the rate of total mass disappearance due to adsorption in the continuity equation.	The value of the additional term is returned and added to Fluent's continuity equation as a source term.	(1)
DEFINE_SOURCE (Energy)	Used for implementing the heat generation & thermal mass contribution terms in the energy equation.	The value of the additional terms is returned and added to Fluent's energy equation as source terms.	(7), (9), (10)
DEFINE_SOURCE (Species)	Used for implementing the rate of mass disappearance due to adsorption of species i in the species conservation equation.	The value of the additional term is returned and added to each of Fluent's species equations as a source term.	(4)
DEFINE_DIFFUSIVITY	Used for calculating the molecular diffusivity and dispersion coefficient.	The value of molecular diffusivity is calculated, stored in a UDMI, and passed to the custom LDF solver to calculate k_i. The value of the dispersion coefficient is calculated and returned to Fluent, overriding the default diffusivity value.	(5), (6)

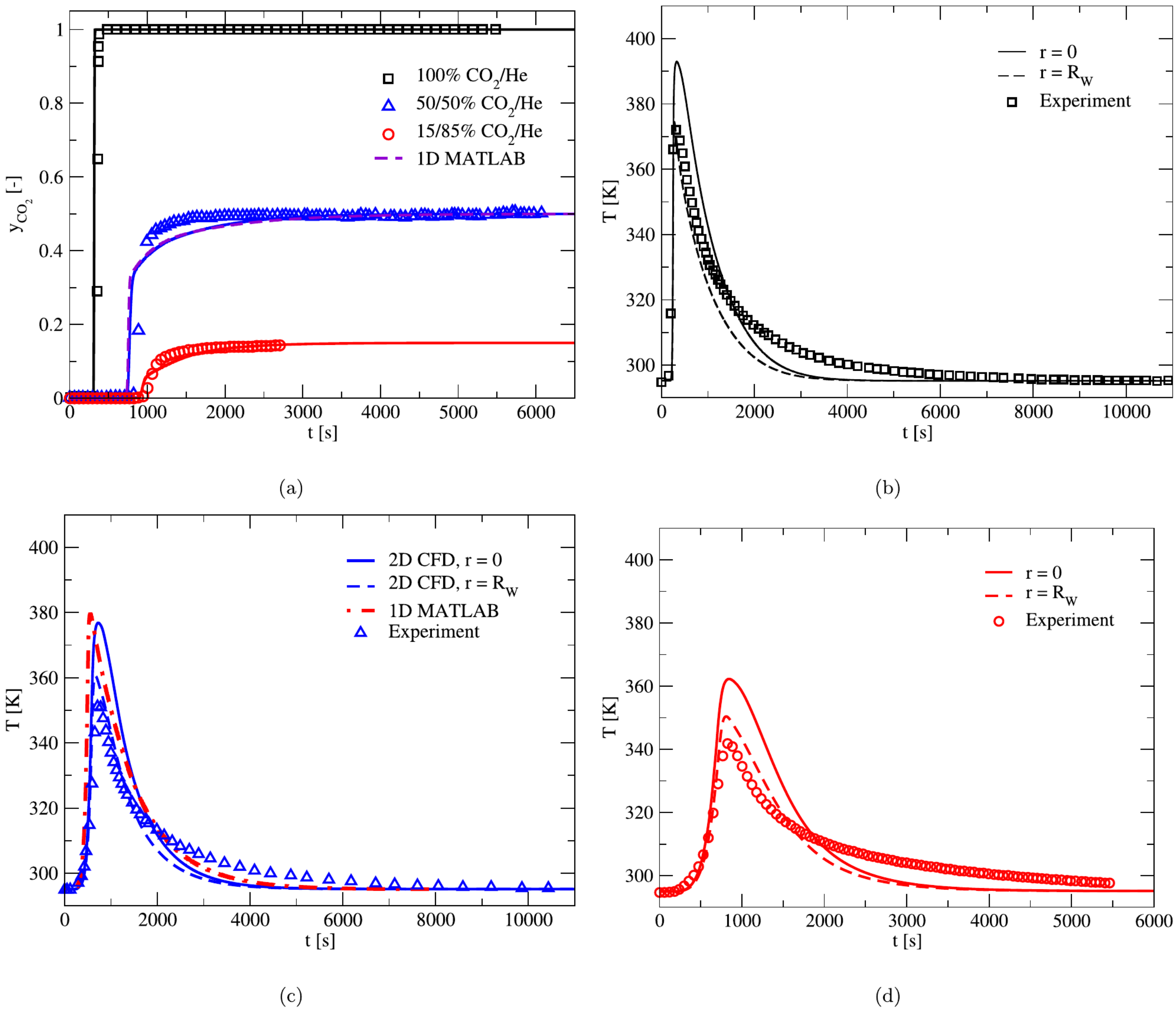

Fig. 6. CO_2 composition and temperature breakthrough curves predicted for different CO_2/He inflow gas compositions. Here, the lines represent the CFD results and the symbols refer to experimental data by Wilkins (Wilkins and Rajendran, 2019), **1D MATLAB refers to data by Wilkins (Wilkins and Rajendran, 2019)**. (a) Molar fraction breakthrough of all 3 compositions. (b) Temperature breakthrough, 100/0% mole CO_2/He case. (c) Temperature breakthrough, 50/50% mole CO_2/He case. (d) Temperature breakthrough, 15/85% mole CO_2/He case.

operating temperatures and pressures for all are 0.96 bar and 23 °C (excepting 25/75%, where T = 22.7 °C). The N_2 isotherm is modelled using the Dual Site Langmuir equation with parameters fitted via the Equal Energy Sites procedure (DSL-EES), as it has reported the best predictions for single-component N_2 breakthroughs in the work by Wilkins (Wilkins and Rajendran, 2019) (see the parameters in Table 5). The result of the three cases is shown in Fig. 5. For comparison we show 1D results from the work by Wilkins (Wilkins and Rajendran, 2019).

As can be seen, the CFD-based results agree well with the experimental points in all three cases. From the composition breakthrough curves, we observe that the model is able to reproduce the breakthrough timing, the front shape, as well as the final composition for each case with reliable accuracy. The 50/50% case exhibits almost perfect accuracy, followed by the 25/75% case and, lastly, the 85/15% case, whose agreement can still be deemed reasonably good. As for the temperature breakthrough, on the contrary, the 85/15% case provides the best match of the three. Granted, the peak is not as tall as in the experiment (which, again, could be attributed to measurement uncertainty); however, it is only different by 1 °C, and the remaining portion of the curve corresponding to the temperature decay matches well the experimental data points. The 50/50% case follows in terms of accuracy, as the temperature tail is slightly higher than in the experiments, but reaches a peak that is only less than 0.5 °C away from the experiments. Finally, the 25/75% case ranks third in temperature breakthrough accuracy, since its tail is about 0.5 °C away from the data points. Still, its breakthrough peak is very similar to that of the experiment. Comparisons between 1D model results and 2D CFD results show good agreement, however, it can be seen that CFD results are closer to experimental data. Additionally, it is clearly seen that the radial temperature gradient exists in the bed, see Fig. 5.

In this way, it was demonstrated that this CFD-based model is capable of reproducing the adsorption column dynamics of a N_2/He pilot experiment successfully. It must be noted that the exact same framework was used for all three cases, where the only changing parameters were the feed composition, flow rate, and ambient conditions of pressure and temperature.

3.4. Single-component adsorption of CO_2

To further assess the capabilities of the model, a whole different set of simulations was performed involving CO_2. Three inlet compositions were tested: 100% CO_2, 50/50% CO_2/He, and 15/85% CO_2/He. These

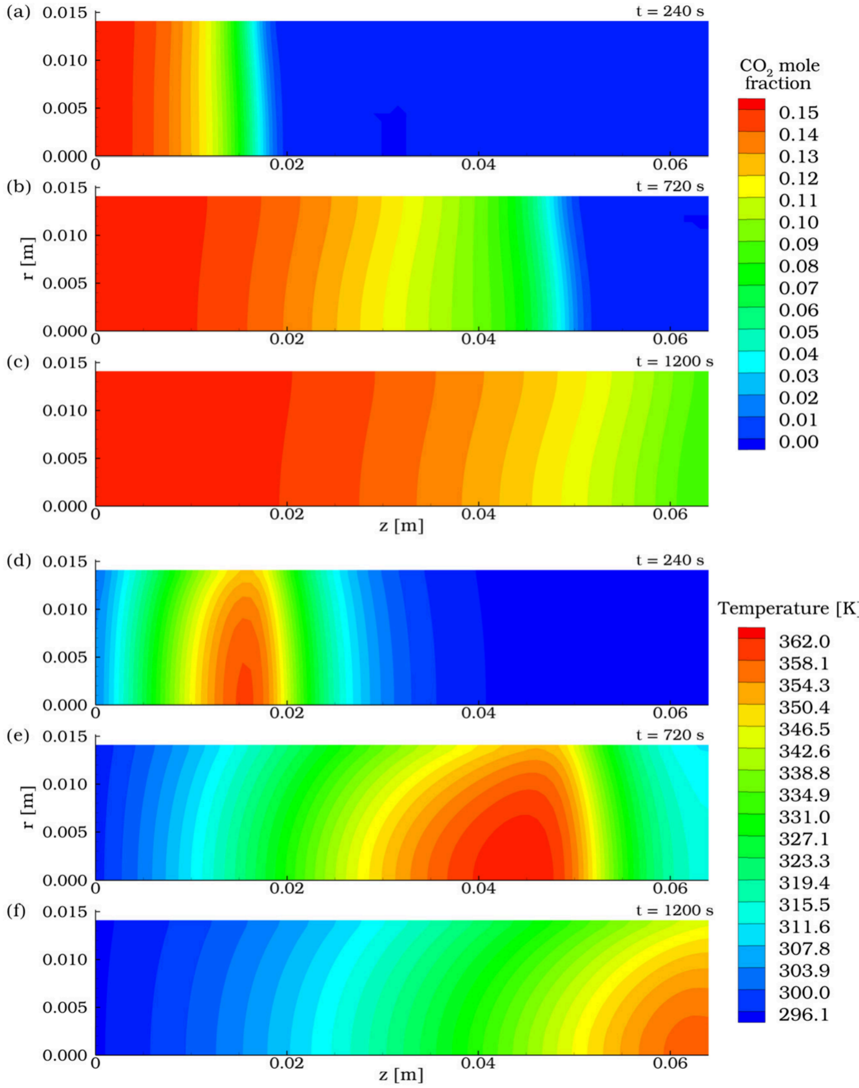

Fig. 7. 2D contours of CO_2 molar fraction and temperature across the bed for the 15/85% CO_2/He case at different times. (a) and (d) $t = 240s$. (b) and (e) $t = 720s$. (c) and (f) $t = 1200s$.

which were reported in the experimental work by Wilkins (Wilkins and Rajendran, 2019). The flow rates are 350, 315, and 700 ccm, respectively, and the operating temperature and pressure are set at around 22 ^{0}C and 1 bar. Since the CFD framework was coded for modelling the binary adsorption of N_2 and CO_2 from the start of this research, the adsorption parameters are already included in the model, and correspond to the Dual Site Langmuir isotherm for CO_2 (refer to the Table 5). The only necessary changes in the model for this set pertain only to the inlet feed, experimental conditions, and the diffusivity coefficient formulation, which must now be calculated between helium (carrier) and CO_2 (adsorbate), rather than He and N_2.

The breakthrough plots are shown in Fig. 6. For comparisons, we show 1D modelling data from the work by Wilkins (Wilkins and Rajendran, 2019). Note that the model features good agreement with all three different experiments regarding the composition breakthrough curve. Unlike the N_2 breakthrough, the CO_2 curve features a sharp front at the start of the breakthrough, which later becomes smeared as the adsorbent saturates. The same tendency is also observed for the 50/50 and 15/85 cases, although to different degrees. This behaviour is characteristic of CO_2 due to its highly non-linear Type I isotherm (as opposed to N_2, whose isotherm is fairly linear). The breakthrough curve is expected to mirror the shape of the isotherm, and our model is able to reproduce it well. With regard to the temperature breakthrough, all the CO_2 cases display a much higher temperature rise compared to N_2 due to its higher equilibrium loading and isosteric adsorption heat. Although our model overpredicts the peak breakthrough temperature, the tendencies still match reasonably well, including the timing. It can be seen that the temperature difference between the axis and the side wall is about 30 K for the 100/0% case, 18 K for the 50/50% case, and about 12 K for the 15/85% case. As the front progresses, this temperature difference decreases. Finally, it should be noted that in spite of a good agreement between 1D model results and 2D CFD, see Fig. 6a and Fig. 6c, 2D CFD model is able to predict the radial temperature gradient in the bed during adsorption processes. This effect is attributed to the no-slip side walls and to the radial heat transfer between the bed and ambient air.

Finally, Fig. 7 shows the two-dimensional contours for the temperature and CO_2 composition for the 15/85% CO_2/He case. Notably, the composition contour does feature important gradients both before and after the breakthrough, as the front does not resemble a flat plug-flow-like profile. This result is of importance, as it shows that, in addition to the clear existence of temperature gradients, composition gradients in the radial direction are also possible, even in a lab-scale system. The temperature contour displays sizeable thermal gradients across the bed at all the times selected. These gradients are not only present during the adsorption regime, but well beyond the breakthrough too (cooling

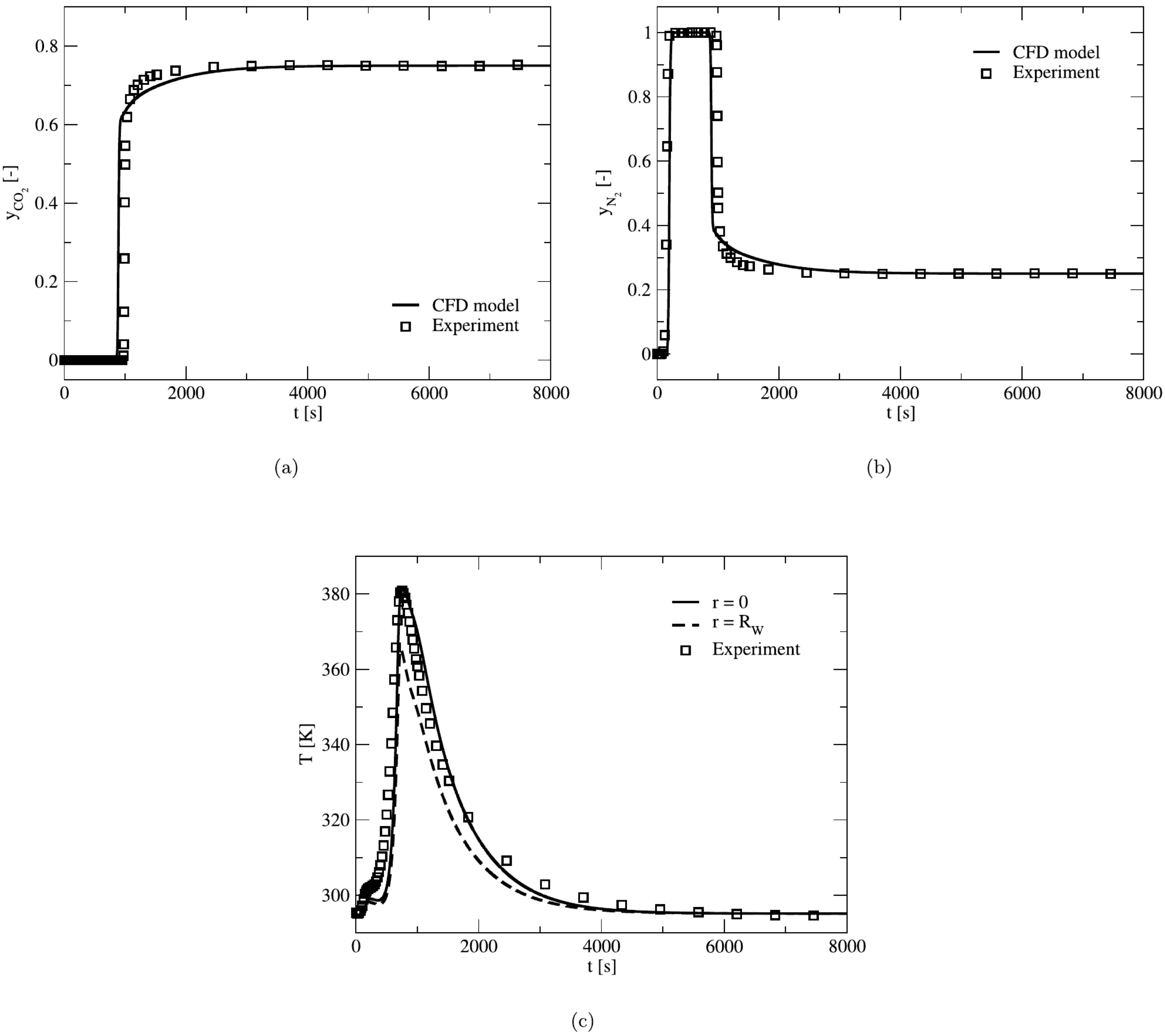

Fig. 8. Composition and temperature breakthrough curves for competitive adsorption between CO_2 and N_2 with He as carrier. Inlet composition: 75/25% CO_2/N_2. The lines represent CFD results, and 'Experiment' refers to the work by Wilkins (Wilkins and Rajendran, 2019). (a) CO_2 composition breakthrough. (b) N_2 composition breakthrough. (c) Temperature breakthrough.

regime). The radial gradients for this case are about 12 ^{0}C, which is a significant quantity that our model is able to predict.

3.5. Competitive adsorption of CO_2 and N_2

The two previous sections explored single-component adsorption, where there is only one adsorbing species (either CO_2 or N_2) accompanied by a carrier, which is assumed to be non-adsorbing (He). In the following section, an alternate scenario will be explored, wherein both adsorbing species coexist within the column and undergo adsorption simultaneously. This is known as competitive or binary adsorption.

The equilibrium and kinetics of competitive adsorption for each component are different, which determines their corresponding loading values and breakthrough times. In the case of CO_2 and N_2, the former saturates the bed to a considerably higher degree compared to the latter, which results in an earlier breakthrough for N_2 (lighter product), whilst CO_2 (heavier product) emerges afterwards. As a result, there are two separate breakthroughs for each component. This is also reflected in the behaviour of the temperature breakthrough with the existence of two temperature peaks: the first corresponding to N_2, and the second corresponding to the heavier CO_2. This competitive behaviour is described by the isotherms of each component, where an appropriate isotherm model and fitting procedure must be used to ensure a proper estimation of the experimental data. Wilkins & Rajendran determined that a Dual Site Langmuir (DSL) model with Unequal Energy Sites (UES) fitting procedure was the most appropriate for describing the equilibrium of competitive CO_2/N_2 experiments, since actual experimental binary loadings are considered to fit the model (Wilkins and Rajendran, 2019). The parameters of the DSL-UES isotherm are included in Table 5, and are used in this work for binary adsorption simulations.

One competitive adsorption experiment from the work of Wilkins (Wilkins and Rajendran, 2019) was reproduced using our CFD adsorption model. A gas mixture containing 75/25% CO_2/N_2 is fed into the column at 202 ccm. The bed is initially saturated with helium. The operating pressure and temperature are 21.6 °C and 0.98 bar. The resulting breakthrough curves are shown in Fig. 8.

As can be seen, the model reproduces the experiment with good agreement for all the breakthrough curves. The breakthrough curve of the heavier product, CO_2, is estimated well in the same fashion as in the single-component CO_2 experiments. As is evidenced by Fig. 8(a), the predicted curve displays the same shape as in the experiments, which –as described before– reflects the non-linearity of the type I CO_2

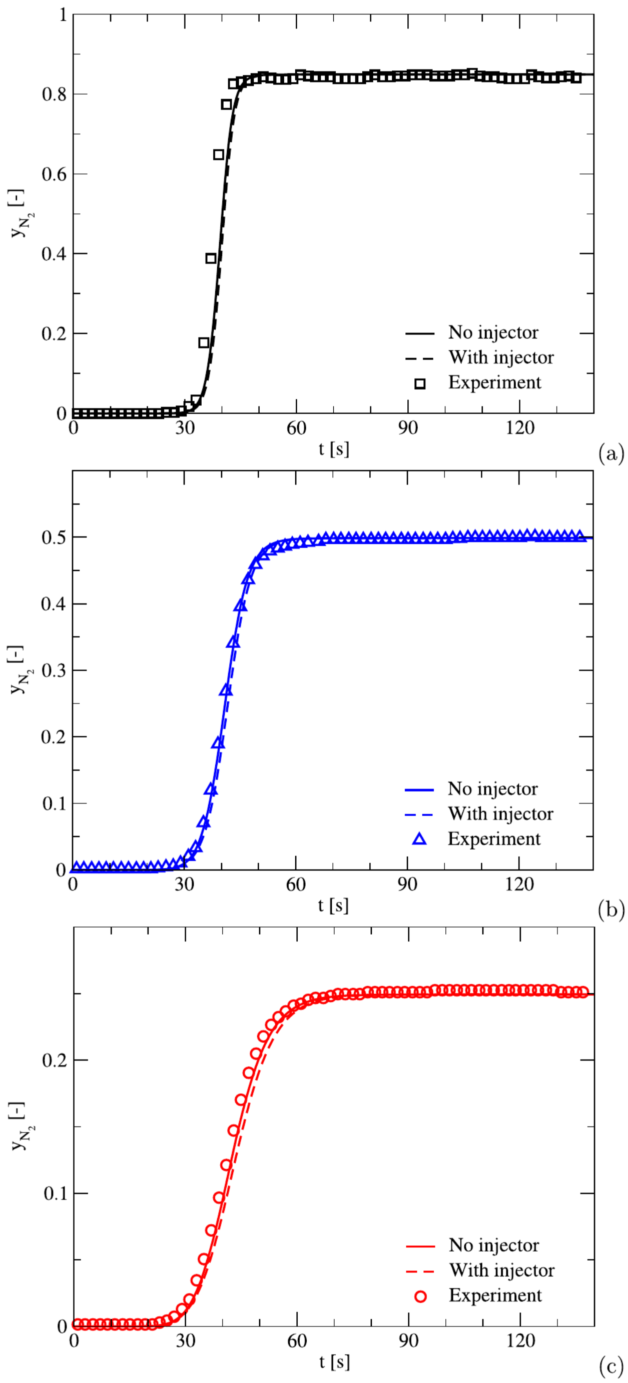

Fig. 9. Comparison of the composition breakthrough curves between the cases with and without an injector, for the three N_2/He compositions. (a) 85/15 N_2/He case. (b) 50/50 N_2/He case. (c) 25/75 N_2/He case. Here, 'Experiment' refers to the work by Wilkins (Wilkins and Rajendran, 2019).

isotherm. Regarding the N_2 composition breakthrough, one feature that the model was able to reproduce accurately was the so-called 'roll-up' effect, which refers to the period of time in which the outlet molar flow of N_2 is greater than its incoming flow rate. This can be observed in Fig. 8(b), where the molar composition of N_2 hits 100% from 200 to 800 s, period during which N_2 is effectively separated from CO_2 due to the relative differences in their adsorption behaviour. When this period is over, CO_2 finally starts to emerge from the column outlet upon having largely saturated the bed. This is how competitiveness is exploited during this separation operation, and our model is able to capture it well.

As for the temperature breakthrough shown in Fig. 8(c), two peaks are observed, which, as mentioned before, correspond to the breakthrough times of each component. The first peak, which happens at around 200 s, corresponds to the heat released by N_2 adsorption. Given that N_2 possesses a relatively low isosteric heat and saturation capacity compared to CO_2, a very short peak results in comparison to the one from CO_2, at around 800 s. The model is able to reproduce both peaks, with particularly high accuracy for the notorious CO_2 peak, where not only the magnitude of the heat rise is predicted well, but also the temperature values during the cooling stage after the breakthrough.

4. Study of 2D spatial effects: inlet injection

It is now time to deepen the exploration of the benefits of higher-dimensional modelling. Unlike uni-dimensional models, we are now able to add geometrical complexity to our CFD domain in order to attain a more accurate resemblance of a real adsorption column. The results presented so far correspond to a rectangular domain, where the intake section is of the same width as the rest of the column. However, in real systems, the feed is injected through a pipe of smaller radius. This injection creates a jet that influences the development of the flow and fronts in the entrance section and beyond, much to the contrary of a rectangular or 1D geometry, where the fronts remain virtually flat. Thus, the next results aim to showcase the capabilities of a 2D model for exploring the spatial effects of a jet injection on adsorption.

These next simulations were based on the same experimental setup as described previously. The sole difference lies in the inclusion of a small injection pipe at the inlet, which is concentric with the rest of the column. The new model domain is shown in Fig. 1(e). Based on our mesh study in Section 3.1, it is determined that a medium mesh best combines performance and accuracy. Therefore, the aforementioned domain is meshed with a similar resolution, resulting in 2.7 thousand cells. The inlet velocity was adjusted to the new inlet pipe maintaining the same flow rate. The inlet pipe is considered to be hollow (no packing, unobstructed flow), and no adsorption takes place there. All other settings in the Fluent case file were kept the same. Apart from specifying the new ID of the porous zone, no changes were made to the adsorption UDF.

In a similar fashion as before, this new model was used to simulate dynamic column breakthrough experiments for the same three inlet N_2/He compositions explored in Section 3.3. As such, we can compare the breakthrough curves with and without injection. This is shown in Figs. 9(a), 9(b), and 9(c), which correspond to compositions 85/15 N_2/He, 50/50 N_2/He, and 25/75 N_2/He respectively.

As can be observed, in all three cases, the inclusion of the injector produces a small shift in the breakthrough curve. The time delay with respect to the no-injector case is approximately 0.5 s for the 85/15% case, 0.7 s for the 50/50% case, and 0.9 s for the 25/75% case. Despite this delay, the inclusion of the inlet injection does not significantly affect the quality of agreement of each case for these lab-scale simulations, since all curves are still very close to the experimental values and follow the same trend and shape as their flat-entrance counterparts. Regarding the temperature breakthrough, the effects of the jet can be deemed negligible for these cases, since only an increase of 0.01 K is observed near the end of the cooling stage. Thus, the corresponding plots have not been included.

The small delay with respect to the flat-profile cases can be attributed to the conjoint effect of the next two factors: First, given that the inlet pipe adds extra length to the domain, the flow must now travel a longer distance. Considering the adjusted new entrance velocity through the small inlet pipe (0.2342 m/s) and the additional distance that the flow must travel (0.0141 m), the delay can be calculated: approximately 0.06 s, for all cases. This means that the longer travel distance is not the primary source of the delay. Instead, a second factor provides a more plausible explanation: the entrance effects of the injection jet. As explained next, these cause the upper section of the en-

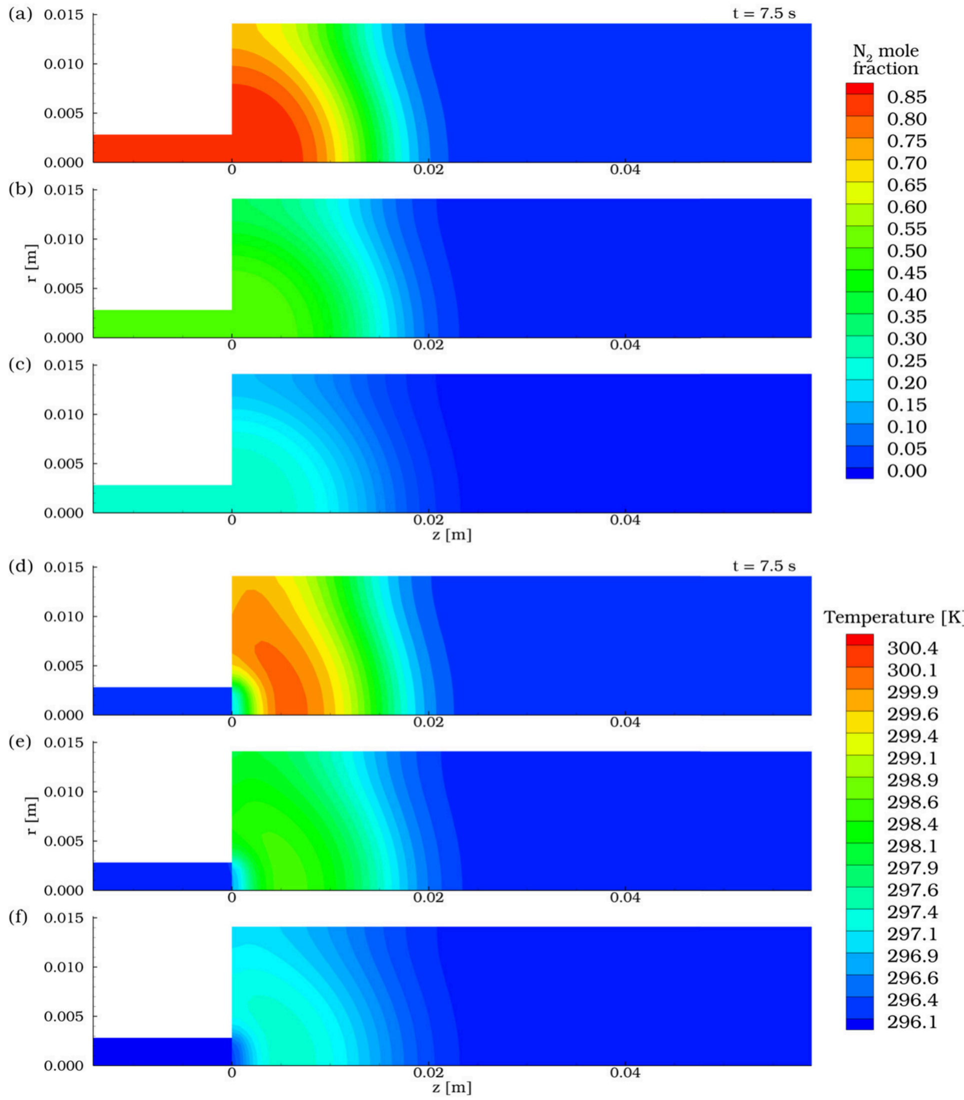

Fig. 10. 2D N_2 molar fraction and temperature contours for different compositions near the start of a DCB experiment ($t = 7.5s$) for the injection case. (a) and (d) 85/15% mole N_2/He. (b) and (e) 50/50% mole N_2/He. (c) and (f) 25/75% mole N_2/He.

trance to be used ineffectively ($r \rightarrow R_W$), which increases the time that it takes for the bed to saturate. In order to gain insight on this, we explore the 2D profiles and radial concentration profiles, which allow us to observe the distribution of matter and heat around the entrance and the evolution of fronts in the remaining length.

First, we look at the distribution of N_2 in the bed at two different times: 7.5 seconds (Fig. 10) and 24 seconds (Fig. 11). At 7.5 seconds, the species fronts are in the early stages of their development. The influence of the injection pipe is evident, as nitrogen first begins saturating the bed around the vicinity of the jet. In the flat-profile case (no jet), the whole region is saturated uniformly, whereas in this new case, the corner furthest away from the jet has yet to be saturated. This can also be observed by examining the radial concentration profiles in Fig. 12, where it is evident that the jet induces a higher concentration near the axis and a lower one around the walls. This explains why the breakthrough curve shifts to the right, as it takes slightly more time for the nitrogen to spread across the low-concentration regions and even out the concentration front. It must be noted that the entrance effects are mostly prevalent during the first half of the column. Past that point, the fronts finish their development and adopt a flat, uniform shape again. This can be seen in Fig. 11(a-c), at 24 seconds.

Nevertheless, as opposed to the composition contour, the effects of the injection pipe on the temperature contour can be significantly more long-lasting. In Fig. 10(d-f) we can appreciate how the jet affects the initial stages of the temperature profile in a similar fashion as in the composition profile, for all compositions. However, unlike the composition contours, at $t = 24s$ the temperature contour still carries the entrance irregularity due to the jet injection, as is seen in Fig. 11(d-f). For reference, compare Fig. 11(d) with Fig. 3(a) at $t = 24s$. There, it is clear that, although the same experiment is reproduced, the presence of an injector accounts for a significant visual difference, manifested later as an alteration to the breakthrough curves –which could be amplified when dealing with large and intricate industrial column designs. These results demonstrate that the capability of a higher-dimension model for implementing realistic geometrical features (such as an injection pipe) is of significant importance when it comes to capturing the effects of said features in an adsorption process, as it helps us attain a more detailed insight into what is actually occurring inside the extents of the domain.

5. Conclusions

As outcome of this research work, a multidimensional CFD-based framework was developed for the 2D and 3D modelling of adsorption column dynamics. The mathematics of the adsorption model were successfully implemented into ANSYS Fluent v19.2 by modifying each of

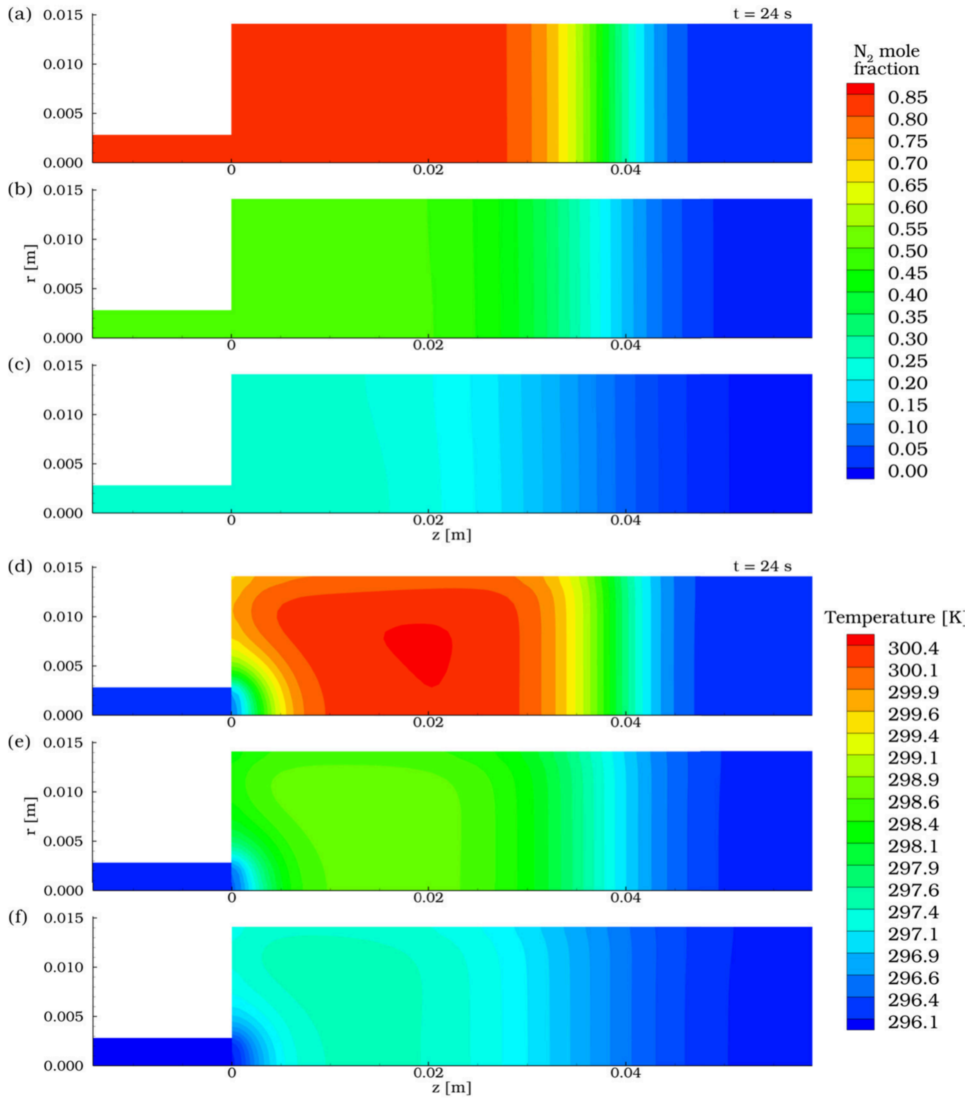

Fig. 11. 2D N_2 molar fraction and temperature contours for different compositions around halfway through a DCB experiment ($t = 24s$) for the injection case. (a) and (d) 85/15% mole N_2/He. (b) and (e) 50/50% mole N_2/He. (c) and (f) 25/75% mole N_2/He.

the existing transport phenomena equations in order to quantify the material and thermal effects of adsorption. Fluent's UDF functionality played a vital role in the implementation of the custom mathematics for adsorption, as it allowed us to code each required transport term as source/sink terms, and to reformulate and implement certain transport properties, such as diffusivity and axial dispersion. Furthermore, a numerical solver for the Linear Driving Force (LDF) differential equation was also coded using UDFs, resulting in a custom 2nd-order implicit discretization scheme that favours high-speed runs, making our model solvable within a reasonable time frame. The model was validated against various Dynamic Column Breakthrough experiments involving both single-component and competitive adsorption of nitrogen and carbon dioxide on zeolite 13X, and was able to reproduce the breakthrough curves well in all cases. The mesh independency and time-step/iterations studies allowed us not only to determine the right balance between speed and accuracy, but also to demonstrate the robustness of our model, as it was able to produce reasonably good results even with the most ambitious configurations (high time steps, coarser mesh). The simulation results of the cases involving CO_2 in their inflow gas compositions showed the existence of significant radial temperature gradients with values as high as 10^3 K/m around the heat propagation front. Finally, this model and the subroutines implemented can be used by researchers to carry out numerical simulations of adsorption columns of different geometries and sizes, such as horizontal fixed beds with vertical injection of gas, where 1D models cannot be used.

CRediT authorship contribution statement

Henry Steven Fabian Ramos: Data curation, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft. **Chinmay Baliga:** Data curation, Investigation, Software, Validation, Visualization, Writing – original draft. **Arvind Rajendran:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Petr A. Nikrityuk:** Funding acquisition, Project administration, Supervision, Writing – review & editing, Formal analysis, Methodology.

Declaration of competing interest

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honorarium; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships,

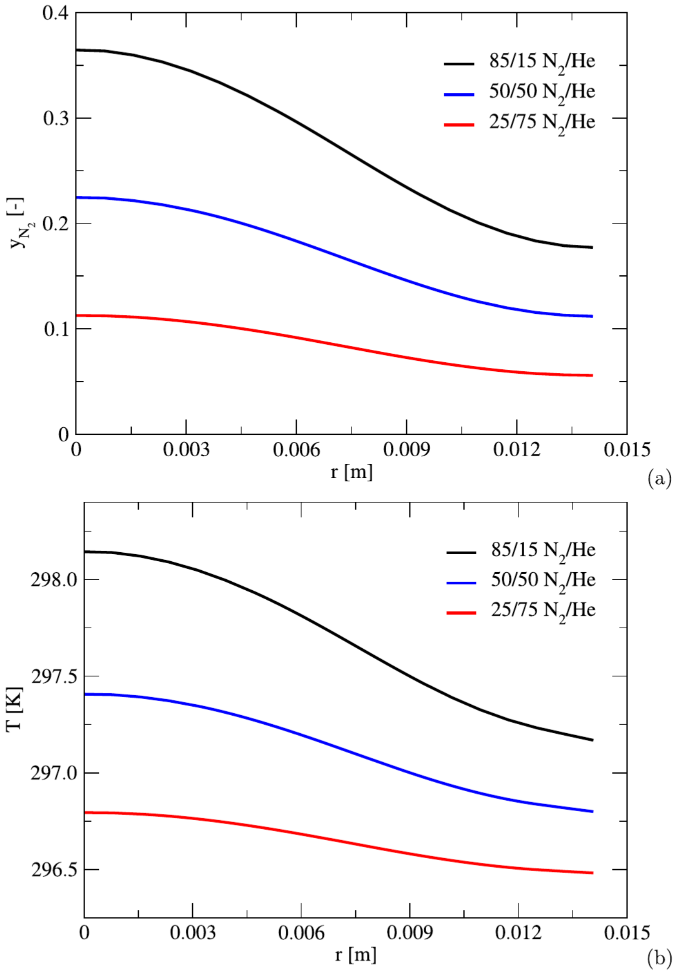

Fig. 12. Radial profiles of composition and temperature for the injection case across $z = 0.015\,m$ at 7.5 s. (a) Molar fraction. (b) Temperature.

affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ces.2023.119606.

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